



Fairer benchmark of group contribution and machine learning models for property prediction: A new data splitting strategy

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ABSTRACT

Accurate prediction of thermophysical properties is important in chemical engineering, where group-contribution models (GCM) have been used extensively. The norm when developing GCMs is to use all available data for parameter estimation, preventing a fair comparison with machine learning (ML) methods that require separate training, validation, and testing data. In this study, we first highlight the detrimental effect of missing groups resulting from using conventional split methods (random and cluster-based) in the development of GCM and ML models using groups as features. This was illustrated by developing property models for the critical point properties (critical temperature, critical pressure and critical volume) as well as the acentric factor. To alleviate this, we propose a novel hybrid splitting algorithm that first ensures that all available groups are represented using the smallest subset of compounds possible and then supplements the subset with molecules based on the Butina clustering of the remaining compounds. The methods show performance close to the "optimal" possible result produced from using all data for the model calibration, and provide a more fair basis for comparing GCM with ML-based methods. We further benchmark the GCM with seven ML techniques (random forest, decision tree, gradient boosting, extreme gradient boosting, Gaussian processes and support vector machines as well as deep neural networks) using groups as features and three graph neural network models (attentiveFP, MEGNet and GroupGAT). The results show that GroupGAT consistently outperforms other methods on the external test dataset, achieving lower errors than both traditional and ML-enhanced GC methods.

1. Introduction

Molecular properties are fundamental in determining the functionality and suitability of a compound in a given application. The nature of these properties is diverse (thermal, environmental, safety and transport-related) and covers a wide range of classes (primary, secondary and functional properties). If one further considers the size of the plausible chemical space and the experimental complexity associated with some of these properties, complete mapping of the chemical-property space is impossible (Reymond, 2015; Ruddigkeit et al., 2012). As such, the development of property models becomes indispensable for chemical engineering practice. These property models can take various forms: compound and property-specific correlations or generalizable models. An example of the property/compound specific models are the Antoine correlation for vapor pressure and the correlations used for heat capacity. Generalizable models are generic models that can extend to "any compound". An example of the latter is

quantitative structure-property relationships (QSPRs). This approach aims to relate the structural information of the molecular in a digitized format, also referred to as a descriptor (usually in a numerical format), to the target property of interest through a mapping function. As such, success in terms of accuracy and applicability, highly depends on the quality and synergy between the three composing elements (descriptor, mapping function and the underlying data used). Descriptors can be general structural information (e.g., number of heavy atoms, number of branches, atom counts, relative surface area, etc.) generated from various cheminformatics software and packages such as Mordred (Moriwaki et al., 2018), PaDEL (Yap, 2011), RDKit (Landrum et al., 2024), etc. Other descriptors are generated through prior knowledge, such as the case of decomposing molecules to fundamental building blocks in the form of substructures and functional groups (or fragments). This representation is often associated with a typical class of QSPRs, namely the group-contribution models (GCM).

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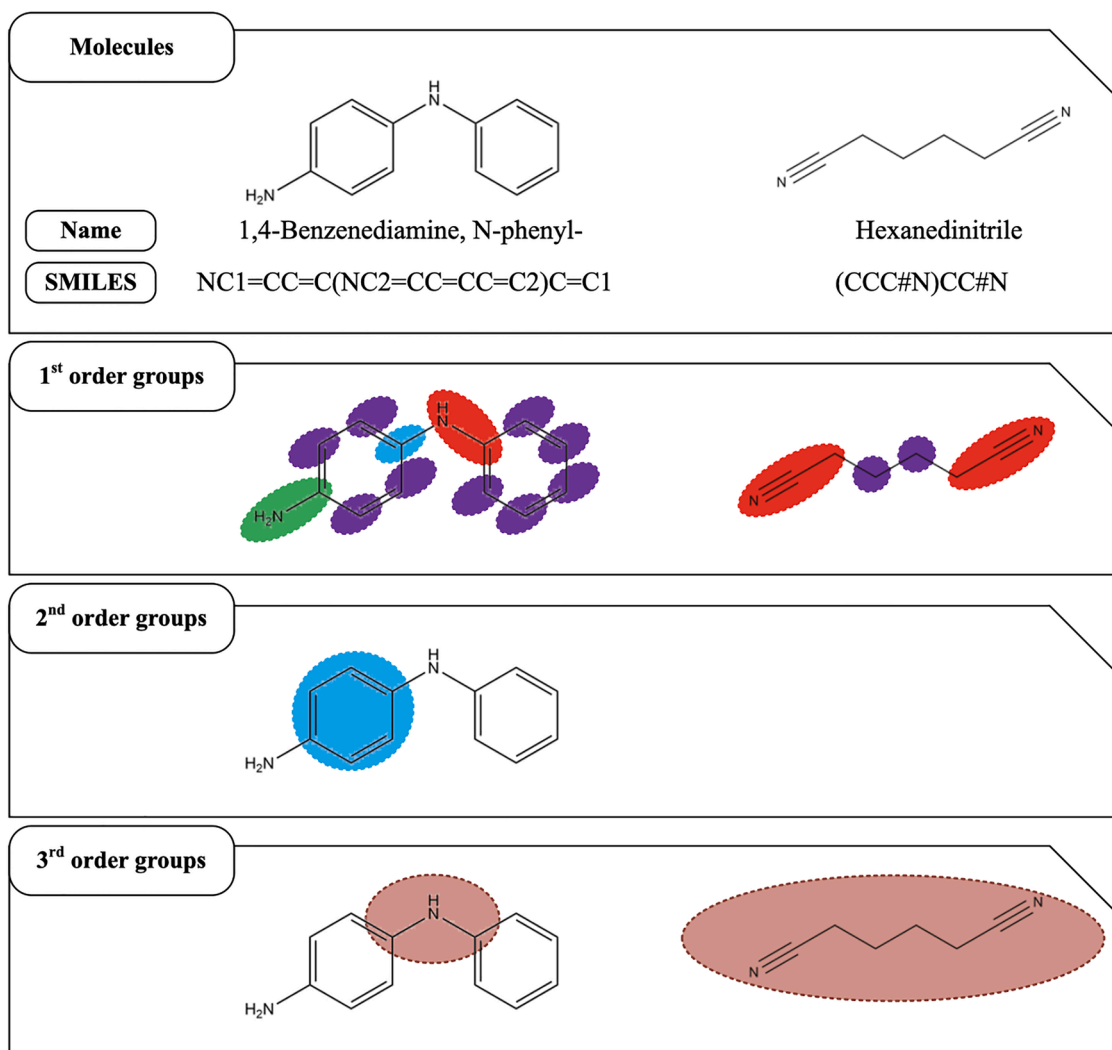


Fig. 1. Example of group identification for two molecules (1,4-Benzenediamine, N-phenyl- and hexanedinitrile) according to the orders of groups defined by Hukkerikar et al. (2012b) and the rules defined by Marrero and Gani (2001). The groups were generated using ICAS ProPred (2024).

1.1. Group-contribution models (GCM)

Group-contribution models (GCM) take as input a vector representing the occurrence of a set of pre-defined groups/substructures. Each group is associated with a property-specific contribution determined through regression. These contributions are linearly added, and the result is transformed in order to calculate the given property. A generic form of a GCM is shown in Eq. (1), where X is the target property, f is a functional transformation with the intent of linearizing the trend of homologous series of the property in question, n_i and c_i are the occurrence and contribution of group i of a total N defined groups. Alternatively, θ is the contribution vector and G is the occurrences matrix.

$$f(X) = \sum_{i=1}^N n_i \cdot c_i = \theta \cdot G \quad (1)$$

While there has been (to some extent) a consensus regarding the functional transformations (f) defined (as these are derived from observing the trend of homologous series), the major difference resides in defining the group segmentation scheme and the approach adopted for identifying the constituent groups of the molecule. This has led to many different GCMs across the chemical engineering literature (Benson and Buss, 1958; Chein-Hsiun Tu, 1995; Constantinou and Gani, 1994; Hukkerikar et al., 2012b; Joback and Reid, 1987; Klinecicz and Reid, 1984; Marrero and Gani, 2001; Nannoolal et al., 2007). Due to their

simplicity, they have been widely used. To mention some of the properties modelled using GCM: critical point properties (Klinecicz and Reid, 1984; Nannoolal et al., 2007), melting point properties (Hukkerikar et al., 2012b; Simamora and Yalkowsky, 1994), flammability limits (Frütiger et al., 2016a,b), solubility parameters (Hukkerikar et al., 2012b; Stefanis and Panayiotou, 2008), etc. One widely used and applied GCM was developed by Hukkerikar et al. (2012a) as an extension of the groups defined by Marrero and Gani (2001). This model utilizes a hierarchical representation of the molecule across three levels with increasing complexity to capture higher-order substructures. Fig. 1 illustrates the segmentation of two molecules according to the rules defined by Marrero and Gani (2001) and the groups defined by Hukkerikar et al. (2012a).

The GCM by Hukkerikar et al. (2012b,a) has been widely applied to 18 pure component thermophysical properties and 22 environmental-related properties. While this approach has been very successful for many properties (critical point properties, normal boiling points, enthalpy of formation, etc.), it has struggled with more complex properties, such as solubility-related properties (octanol/water partition coefficient, Hansen and Hildebrand solubility parameters and enthalpy of fusion). An important note regarding the development of these models is that they typically leverage all available data for model calibration. While some argue that this approach is acceptable due to the parametric nature of classical GC models, which should limit the risk of

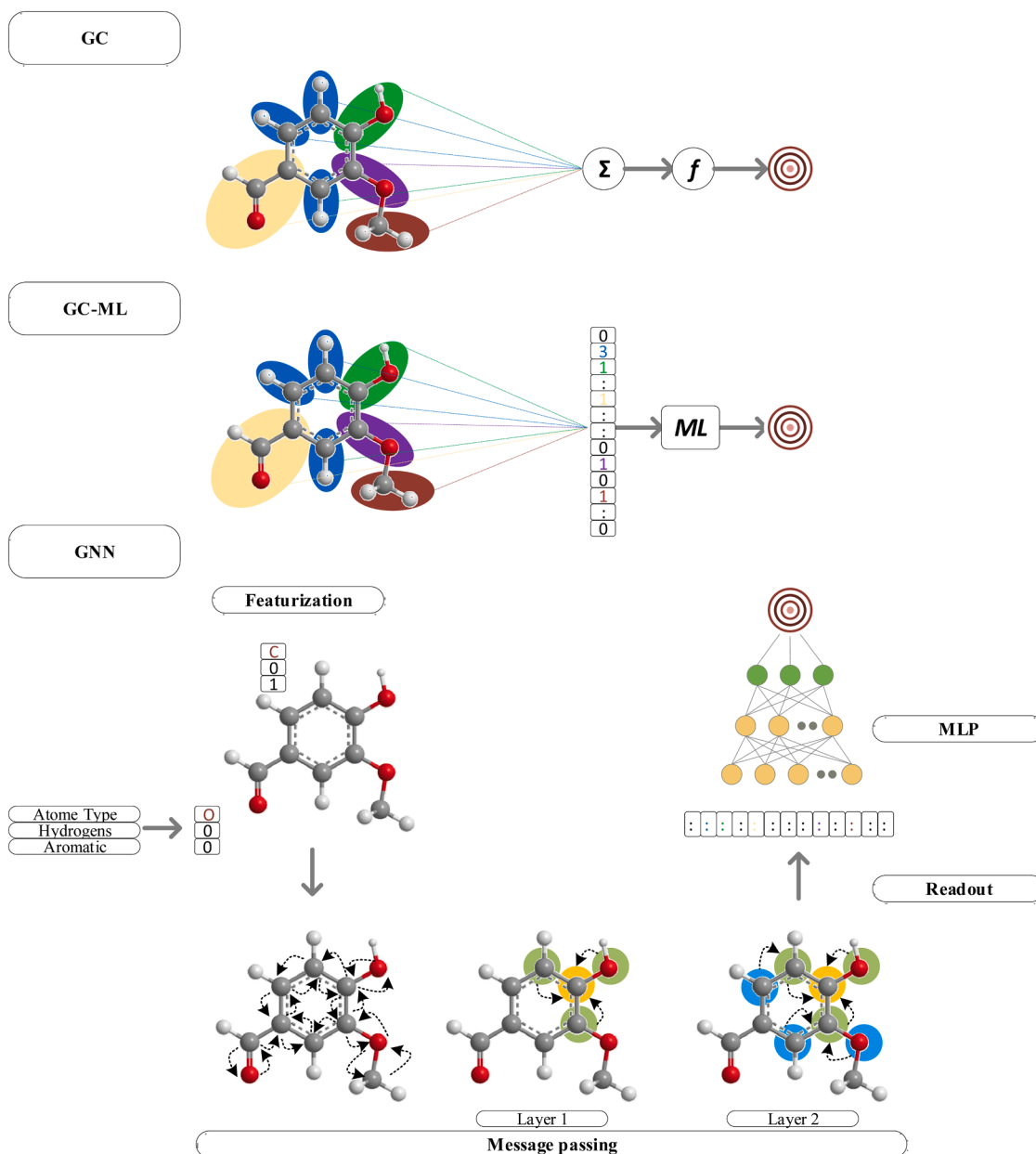


Fig. 2. Schematic of the prediction process of group-contribution models (GCMs), machine learning enhanced group-contribution models (GC-ML) and graph neural networks (GNNs).

overfitting, the lack of validation data prevents a proper assessment of their predictive capabilities on unseen chemical compounds. Although the predictive capabilities might be impaired due to the data used for parameter fitting, which might be erroneous, it is also important to consider some of the drawbacks related to the method itself. For example, the linear nature of the GCMs might hinder them from capturing various nonlinearities associated with certain complex properties. Another potential issue is the fact that the method does not account for potential proximity effects, which influence the contribution of a given group. Furthermore, the rules used to define and segment molecules might not be universal or appropriate for the given molecule or property. While some groups are also rooted in a fundamental understanding of the chemical species and their properties, some have been defined out of convenience (Hukkerikar et al., 2013). This is especially true when using higher-order groups. This is illustrated by the example of hexanedinitrile shown in Fig. 1, where the third-order group encompasses the molecule in its entirety. This could potentially result in

groups with limited presence. Although this might not seem problematic at first, it does result in “error modelling” leading to the “absorption” of the errors, producing predictions with a perfect fit.

Consequently, skewed and overly optimistic performance metrics could be produced. While rectifying the latter would require an overhaul of the specific groups defined and the schemes used to identify them, other attempts to improve GCM have included more systematic outlier deletion techniques (Aouichaoui et al. 2022a; Frutiger et al. 2016a), and/or replacing the linear aspect of the models using machine learning (ML).

1.2. Machine learning enhanced group-contribution models (GC-ML)

Machine learning has heavily influenced many disciplines within the chemical engineering field, from reaction engineering (Mann and Venkatasubramanian, 2023), mechanism identification (Chakraborty et al., 2021), as well as thermodynamics, including prediction of adsorption

isotherms (Chakraborty et al., 2024) and property prediction. Machine learning-enhanced group-contribution models (GC-ML) seek to replace the linear combination of group contributions and the functional transformation (f) with various ML algorithms, including but not limited to Gaussian process regression (GPR), regression trees (RT), artificial neural networks (ANN) and support vector regression (SVR). The rationale behind this is that these models are capable of extracting latent patterns/relations between the input (features) and the desired target, including nonlinearities. In the following paragraph, we introduce a few studies highlighting this line of research. Gharagheizi et al. (2011) combined groups with an artificial neural network (GC-ANN) to model the enthalpy of combustion. The model achieved very high accuracy (less than 0.16% average relative error (ARE)) compared to GCM (ARE of 0.51%) (Frutiger et al., 2016b). Li et al. (2020) developed a GC-ANN for the simultaneous prediction of cetane and research/motor octane numbers (CN, RON, MON). A major benefit of the proposed model is that it leverages the shared interrelation between the properties as a data augmentation strategy to achieve better performance (Li et al., 2020). Aouichaoui et al. (2021) compared three models for mapping the groups to the target properties and their associated uncertainties for the lower flammability limit. The study included a probabilistic neural network (PNN), a Gaussian process model, along with the conventional GCM. Using k-fold cross-validation showcased that GCM performed worse than the two other models with respect to the performance on the external validation set, with GC-GPR performing best (Aouichaoui et al., 2021). Al et al. (2018) used a radial basis function network (RBF), a regression tree and an artificial neural network (feed-forward, single layer) to predict a series of environmental properties consisting of the ozone depletion potential (ODP), the global warming potential (GWP) and the *Daphnia magna* lethal concentration (LC50, 48 hr.). The study showed that GC-ANN and GC-RBFN performed better than GCM across all properties. It also showed that the performance of GCM degrades significantly when applying cross-validation, which could indicate issues when extrapolating to unseen compounds. However, it is important to note that this type of splitting is not suitable when working with GC-based models due to the risk of creating folds where groups are not present, and as such, the GC-based approaches should, in principle, not be used to predict compounds with such groups.

Based on the previous analysis, it is clear that the integration of ML algorithms can lead to an improvement in the predictive performance of the GC-based models, which is also supported in many other studies (Alshehri et al., 2022; Lee et al., 2012; Lee and Lee, 2012). Consequently, ML algorithms could potentially alleviate the bias associated with defining groups and bypass the expert knowledge required to derive such functions. Among such techniques are graph neural networks (GNNs).

1.3. Graph neural networks

Graph neural networks (GNNs) are specialized deep learning (DL) architectures that utilize deep neural networks to extract a latent representation of the molecules from lower-order molecular information. This lower-level information consists of an embedded (or featurized) molecular graph where the nodes and edges are described through a feature vector containing relevant information about the atoms and chemical bonds, respectively. Through an end-to-end learning process, GNNs link the feature extraction to the prediction task, producing a representation that is tailor-made for the property in question.

The information embedded in the graph is iteratively updated through the message-passing mechanism (Corso et al., 2024; Gilmer et al., 2017). At every message-passing layer, the nodes receive incoming messages/information from their neighboring nodes as a set of vectors. These are aggregated into a single vector and used to update the latent representation of the node. In general, the operations of the GNNs can be formalized through three operations: message construction function (an operation that uses neighboring atom information to

construct a vector), update function (a function that depicts how the constructed message should alter the latent representation of the current node) and readout (an operation that converts the graph into a vector to be fed into a deep neural network, DNN (the regressor)). A schematic illustrating the workings of GCM, GC-ML and GNNs can be seen in Fig. 2 using Vanillin as an example.

Recent years have given rise to a wide range of GNNs where the main difference lies in the arithmetic operations defining the message-passing (Chen et al., 2019; Gilmer et al., 2017; Louis et al., 2020; Withnall et al., 2020; Xiong et al., 2020; Yang et al., 2019; Zhang et al., 2021). These models have emerged as a universal ML-based approach for property prediction and have been applied to a wide range of properties relating to chemical absorption, distribution, metabolism, excretion, and toxicity (ADMET) as well as various thermodynamic, transport and safety-related properties. These models have been used to model the flash point (Sun et al., 2020), the lipophilicity and the aqueous solubility (Tang et al., 2020), the critical point properties (Biswas et al., 2023), etc. A further demonstration of their usability is that they are easily extendable to properties that are temperature dependent, as demonstrated by the application of GNNs for the direct prediction of 13 pure component properties covering heat capacities, vapor pressure, enthalpy of vaporization, thermal conductivity and various densities (Aouichaoui et al., 2023a). Such models have also been extended to calculate mixture properties such as infinite dilution activity coefficients (Medina et al., 2022). However, one persisting issue is the lack of interpretability and explainability of such models due to their non-parametric nature (the parameters of the models do not carry any physical meaning). Various techniques have been proposed to elucidate some of the insights captured by the models, which mainly utilize feature perturbation and sensitivity analysis, integrated gradient methods or the attention mechanism, among others (Jiménez-Luna et al., 2020). Our previous study aimed to provide these insights by combining the 1st order groups defined by Hukkerikar et al. (2012b) and the attention mechanism in order to highlight groups of importance to the given property (Aouichaoui et al., 2023c). The attention mechanism provides a score (learned by the model) for all groups present in the molecule. For selected case studies, these were consistent with a chemist's understanding and prior knowledge. To showcase the wide applicability of the model, a follow-up study applied the model to a total of 30 different pure component properties, achieving very high accuracy, especially for properties on which GCM has historically struggled (enthalpy of fusion, aqueous solubility, Hansen/Hildebrand solubility parameters) (Aouichaoui et al., 2023b). For many other properties, the performance was somehow comparable. However, it is important to note that the norm is to use all available data for fitting the GCM.

Directly comparing GNNs, GC-ML models, and GCMs is not straightforward due to a fundamental difference in their development. In particular, ML approaches require external datasets for validation and testing, while GCMs have traditionally, albeit not necessary, used all available data for parameter fitting. The latter is based on the assumption that their parametric nature prevents overfitting. However, this assumption has been scrutinised due to the lack of external validation, which is necessary in the era of ML for ensuring scientific rigor. Additionally, common data splitting techniques such as random splitting may disadvantage GCMs, as they may result in training sets that lack certain groups, hindering their predictive ability.

To address this challenge, we compare a set of splitting techniques and propose a new data splitting algorithm for GC model development called "minimal coverage" or "minimal covering" split. This algorithm identifies the smallest possible training set that contains all available groups in the dataset, ensuring a balanced and fair evaluation of GCMs, GC-ML models, and GNNs. We demonstrate the effectiveness of this approach when supplemented with other splitting methods by benchmarking the performance and uncertainties of these models on predicting critical properties and the acentric factor, properties that are highly relevant in chemical engineering. Furthermore, we investigate

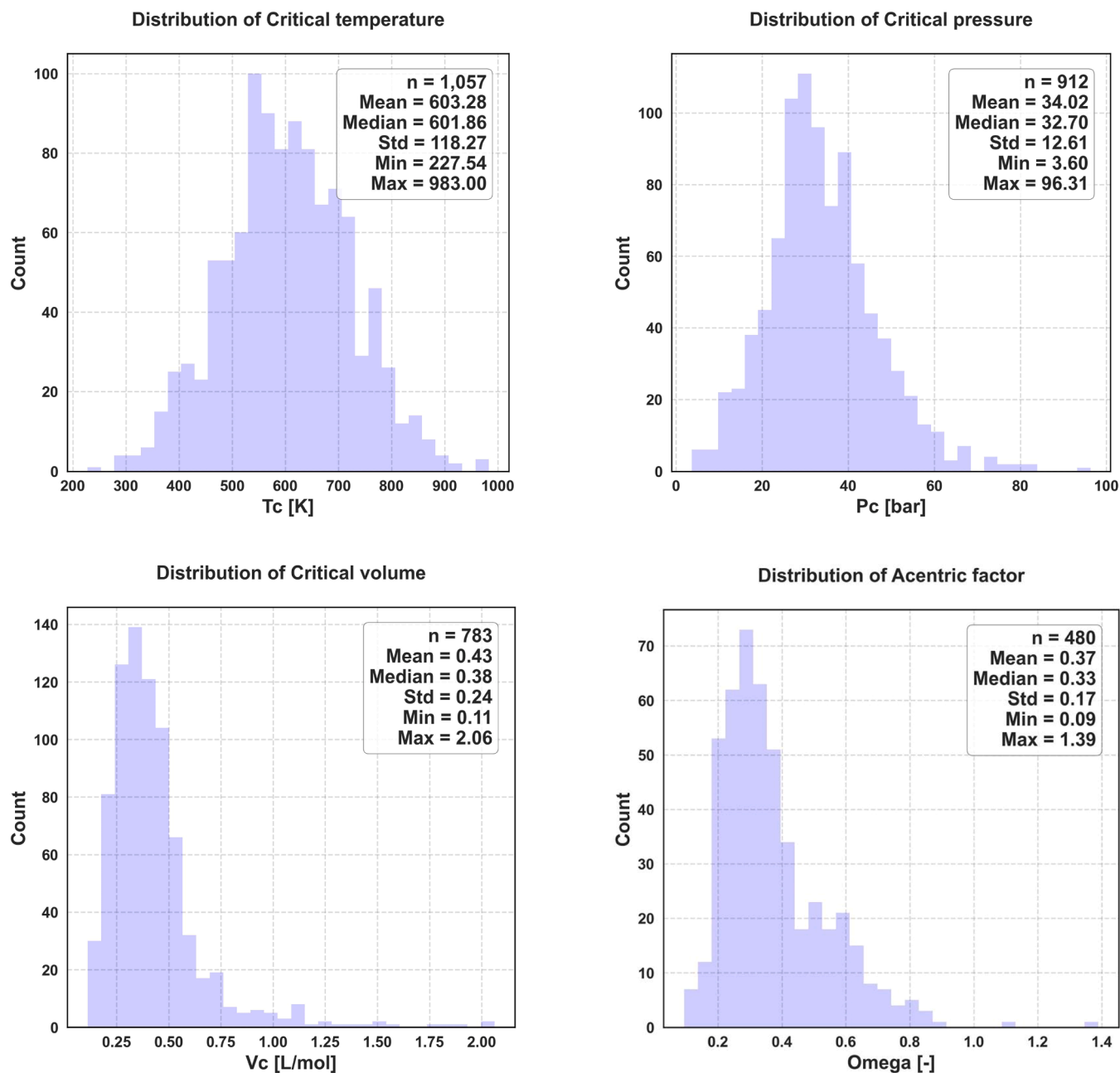


Fig. 3. Distribution and summary statistics of the target values for the critical temperature (T_c), critical volume (V_c), critical pressure (P_c) and the acentric factor (Ω or ω). For the summary statistics: n is the size of the dataset, std is the standard deviation, while min and max are the minimal and maximal values exhibited by the dataset.

the effect of the data splitting and missing features on the various modelling approaches.

2. Methods

2.1. Data

In this study, we focus on the critical properties (including critical temperatures (T_c), pressure (P_c), and volume (V_c)) and the acentric factor (ω) of organic fluids. The choice of properties is largely influenced by their importance in various equations of state (EOS) such as Peng-Robinson and the Soave-Redlich-Kwong EOSs (Peng and Robinson, 1977; Soave, 1972). Additionally, these properties feature heavily in the literature and a wide range of GCMs have been developed for predicting them (Chein-Hsiun Tu, 1995; Constantinou and Gani, 1994; Hukkerikar

et al., 2012b; Joback and Reid, 1987; Klincewicz and Reid, 1984; Marrero and Gani, 2001; Nannoolal et al., 2007; Tahami et al., 2019). The data consists of purely experimental data, primarily collected from the literature (Haynes, 2014; Kleiber and Joh, 2010; Linstrom, 1997; Perry and Green, 2008; Yaws, 2009). The data was previously collected, organized and released as part of another study also dealing with the critical point properties (Biswas et al., 2023). In this work, we only consider molecules with the following atoms: C, H, N, O, Cl, I, Br, F, S and featuring at least two heavy atoms.

Fig. 3 shows the distribution of the target values along with some statistical summary of each dataset. While the T_c presents a somewhat balanced normal distribution-like shape, the remaining dataset exhibits some rather extreme values, which are mostly visible for V_c and ω .

To gain further insights into the chemical species present in the dataset, the distribution of chemical classes is shown in Fig. 4. The

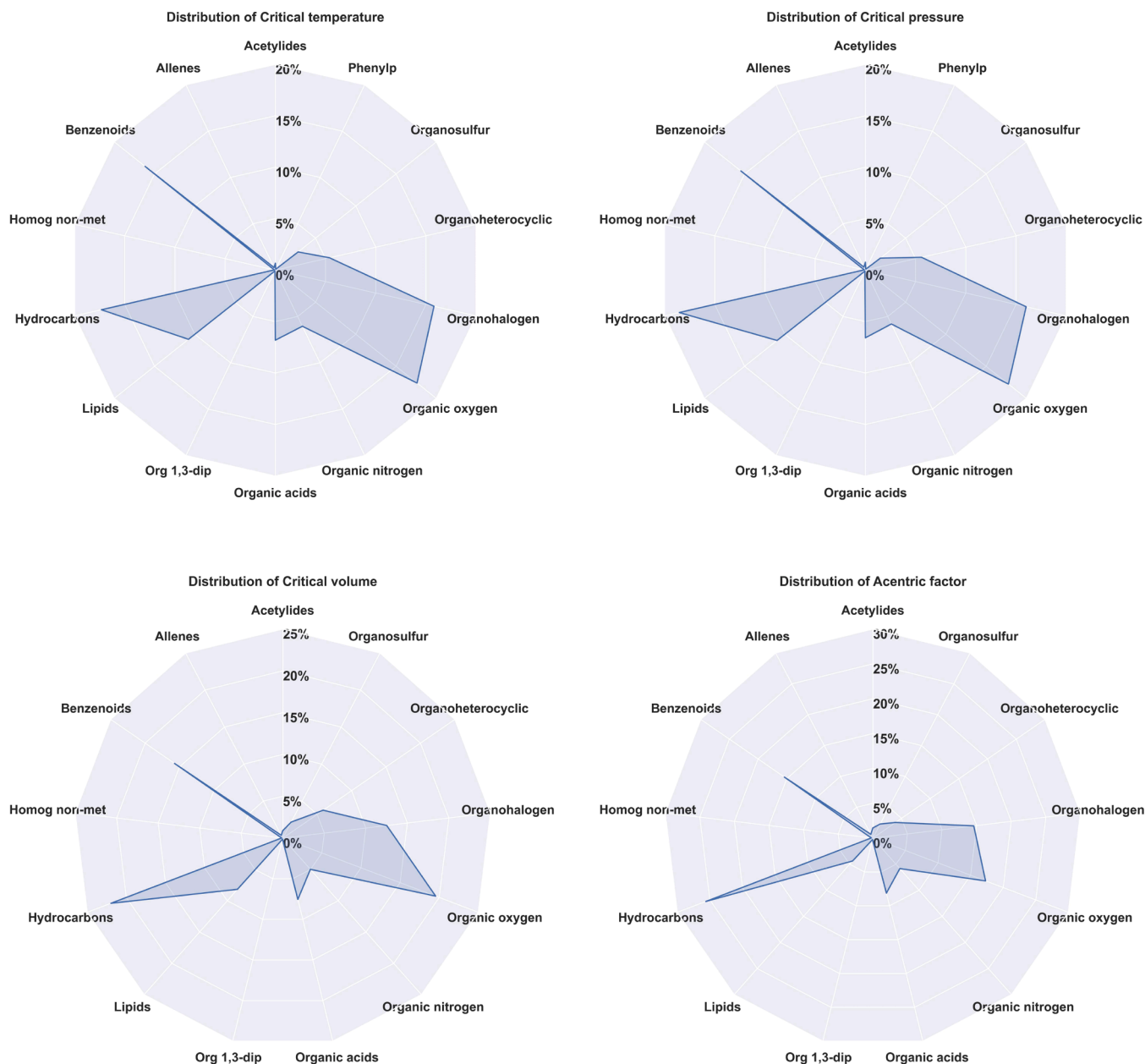


Fig. 4. Distribution of chemical classes present in the dataset. The classes include: Acetylides, Allenes, Benzenoids, homogeneous non-metal compounds, hydrocarbons, lipids and lipid-like molecules, organic 1,3-dipolar compounds, organic acids and derivatives, organic nitrogen compounds, organic oxygen compounds, organohalogen compounds, organoheterocyclic compounds, organosulfur compounds, phenylpropanoids and polyketides.

classes were generated using the open-source tool “*classyfire*” for chemical classification (Djoumbou Feunang et al., 2016). Not surprisingly, all datasets exhibit the same distributions with the data primarily dominated by hydrocarbons, organic oxygen compounds, benzoids and organohalogen compounds. This is to be expected since these datasets are of interest to the same community, are of the same nature and relate to the same condition and phenomenon. This overview also provides an idea on the general domain of applicability of the models developed in this study. It is clear that the present dataset seems to be imbalanced, with some classes not well represented.

2.2. Data splitting: minimal covering split

An important part of developing ML-based models is data splitting. There are many different splitting strategies, each serving its own purpose. Some of the most used strategies include:

- Random splitting: data is split randomly into three sets based on a split ratio.
- Cluster-based splitting: molecules are clustered based on some similarity measures and each cluster is sampled based on a ratio, or the entirety of a cluster is assigned to one of the three sets.
- Scaffold-splitting: data is split according to a scaffold (the core sub-structure), producing three structurally diverse sets.

The generated splits are not insignificant for the performance of the models. Some studies have showcased the significant degradation of the performance of GC-based models when used in ML-based workflows, especially during data splitting (Al et al., 2018; Aouichaoui et al., 2021). This was hypothesized to be because some splits would result in missing out on groups necessary for the prediction. In this work, we introduce the “minimal covering” or “minimal coverage” algorithm, which is designed to find the smallest possible set of molecules that collectively

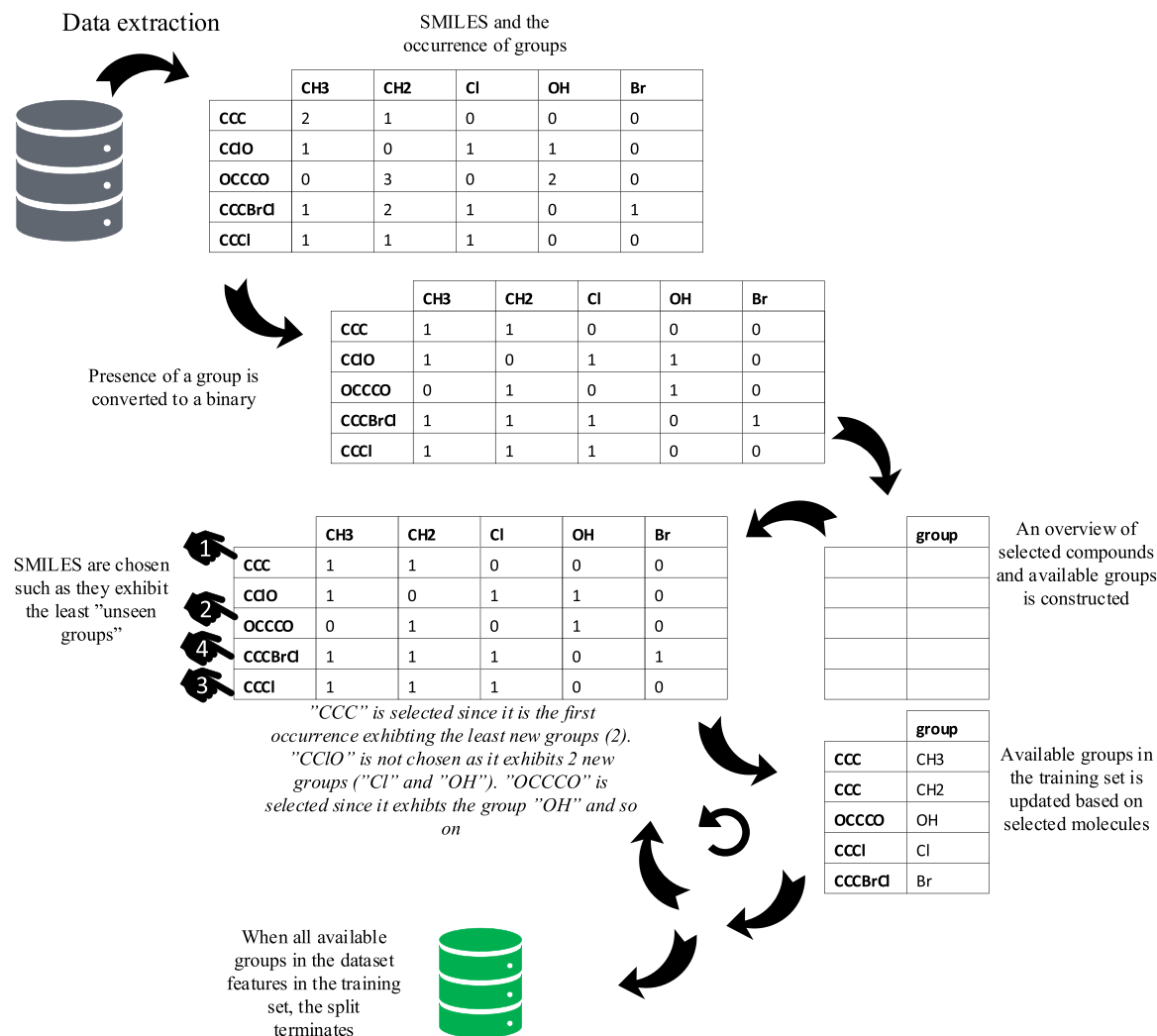


Fig. 5. Schematic of the workflow of the minimal covering/coverage split.

cover all features (groups) present in a dataset. A schematic of the workflow of the split procedure can be seen in Fig. 5.

The algorithm works as follows: an occurrence matrix is first converted into a binary feature matrix, where "1" indicates the presence of a group and "0" indicates its absence. Using a greedy approach, the algorithm iteratively selects molecules that cover the least number of new, uncovered features at each step. This procedure continues until all features are covered or no more features can be covered. The strategy aims to minimize redundancy by preferring molecules that add the fewest new features. The algorithm outputs both the selected set of SMILES and a detailed mapping of which features are covered by each selected molecule.

While this will ensure that all groups are present in the training set, it is important to note that this will result in the smallest possible training set. This might create some challenges for ML-based techniques that rely on features other than groups (e.g., GNNs). In this work, we complement the "minimal covering" split with the Butina clustering method (Butina, 1999). The method clusters molecules based on their Tanimoto similarity with a cut-off of 0.85. The aim is then to sample uniformly from each cluster subject to a final predefined split ratio (in this work, a split ratio of 0.7:0.15:0.15 for training, validation and testing, respectively, is used). For smaller clusters (with one or two members), the training set is prioritized, for clusters of three and four, one data point is allocated to each of the validation and test sets, while the remaining clusters are uniformly sampled according to the defined split ratio. The largest

clusters are sampled according to a re-calculated split ratio based on the sampling from the previous clusters to ensure the final split meets the predefined split ratio.

2.3. Group-contribution modelling

In this work, the groups and functional transformation considered are those defined by Hukkerikar et al. (2012b). The method defines 424 groups: 220 first-order, 130 second-order and 74 third-order groups. The functional transformations used to calculate the properties are shown in Eqs. (2)–(5) for T_c , P_c , V_c and ω respectively and where T_{c0} , P_{c0} , P_{c1} , V_{c0} , ω_0 , ω_1 and ω_2 are constants.

The critical temperature:

$$f(X) = \exp\left(\frac{T_c}{T_{c0}}\right) \quad (2)$$

The critical pressure:

$$f(X) = (P_c - P_{c0})^{-0.5} - P_{c1} \quad (3)$$

The critical volume:

$$f(X) = V_c - V_{c0} \quad (4)$$

The acentric factor

Table 1

Summary of available groups in the various datasets compared to the total defined number of groups by Hukkerikar et al. (2012b).

Property	First-order	Second-order	Third-order
Tc [K]	122/220*	73/130**	27/74***
Pc [bar]	118/220*	69/130**	26/74***
Vc [L/mol]	114/220*	65/130**	22/74***
ω [-]	104/220*	46/130**	15/74***

* Method of Hukkerikar et al. (2012b) defines a total of 220 first-order groups

** Method of Hukkerikar et al. (2012b) defines a total of 130 second-order groups

*** Method of Hukkerikar et al. (2012b) defines a total of 74 third-order groups

$$f(X) = \exp\left(\frac{\omega}{\omega_0}\right)^{\omega_1} - \omega_2 \quad (5)$$

Table 1 shows one of the main limitations of using groups as a molecular descriptor by showing the number of available groups in each order of complexity. As can be seen, for all properties, barely 50% of possible groups are available at the first- and second-order levels, while it is below 35% availability of the third-order groups. In principle, GC methods cannot describe species containing groups not present in the dataset, also known as the missing group problem. This highlights a major limitation on the domain of applicability. While some studies

suggested approaches to predict missing contributions, these have not had wide application or success due to the low predictive accuracy and, as such, this remains an unsolved challenge (Gani, 2019; Hukkerikar et al., 2012b).

The procedure for parameter fitting of the GCM can either be done stepwise or simultaneously, and is described in detail in the work by Aouichaoui et al. (2022a,b), Frutiger et al. (2016b), and Hukkerikar et al. (2012b). An initial guess of the parameters is first generated by leveraging the linear nature of the GCM. Afterwards, the parameters of each level are estimated in a stepwise sequential manner. In this work, we only consider stepwise model fitting.

2.4. Machine learning enhanced group-contributions models (GC-ML)

In this work, we replace the linear part of the GCM with an ML-based model. The governing equation is thus as described in Eq. (6), where X is the property, G is the occurrence matrix and f is a machine learning function being either: decision tree (DT), random forest (RF), support vector regressor (SVR), Gaussian process regressor (GPR), gradient boosted trees (GB), extreme gradient boosted trees (XGB) or a multi-layer perceptron (MLP), also known as deep feed forwards neural networks.

$$X = f(G) \quad (6)$$

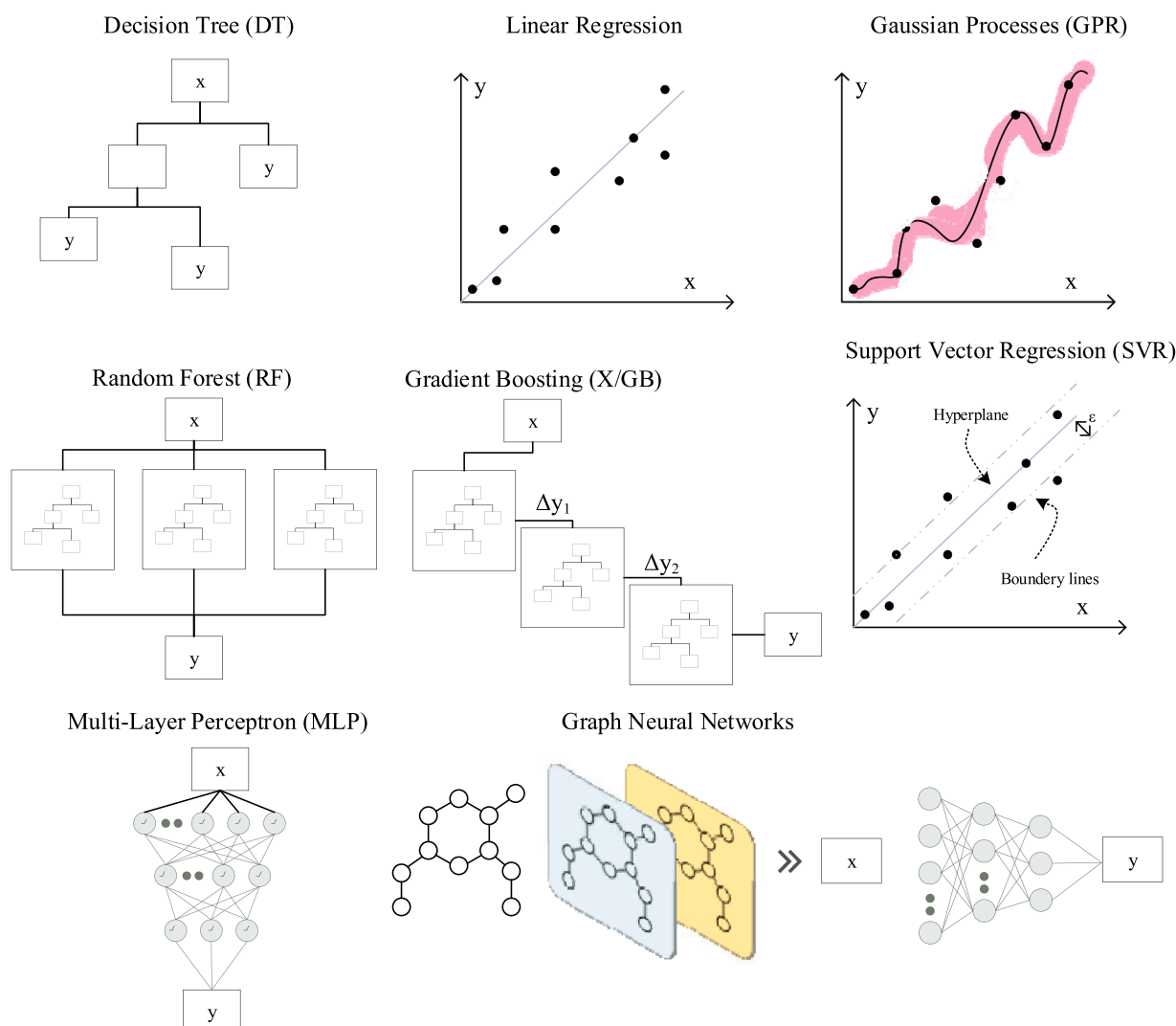


Fig. 6. Overview of the various modelling techniques used in this study.

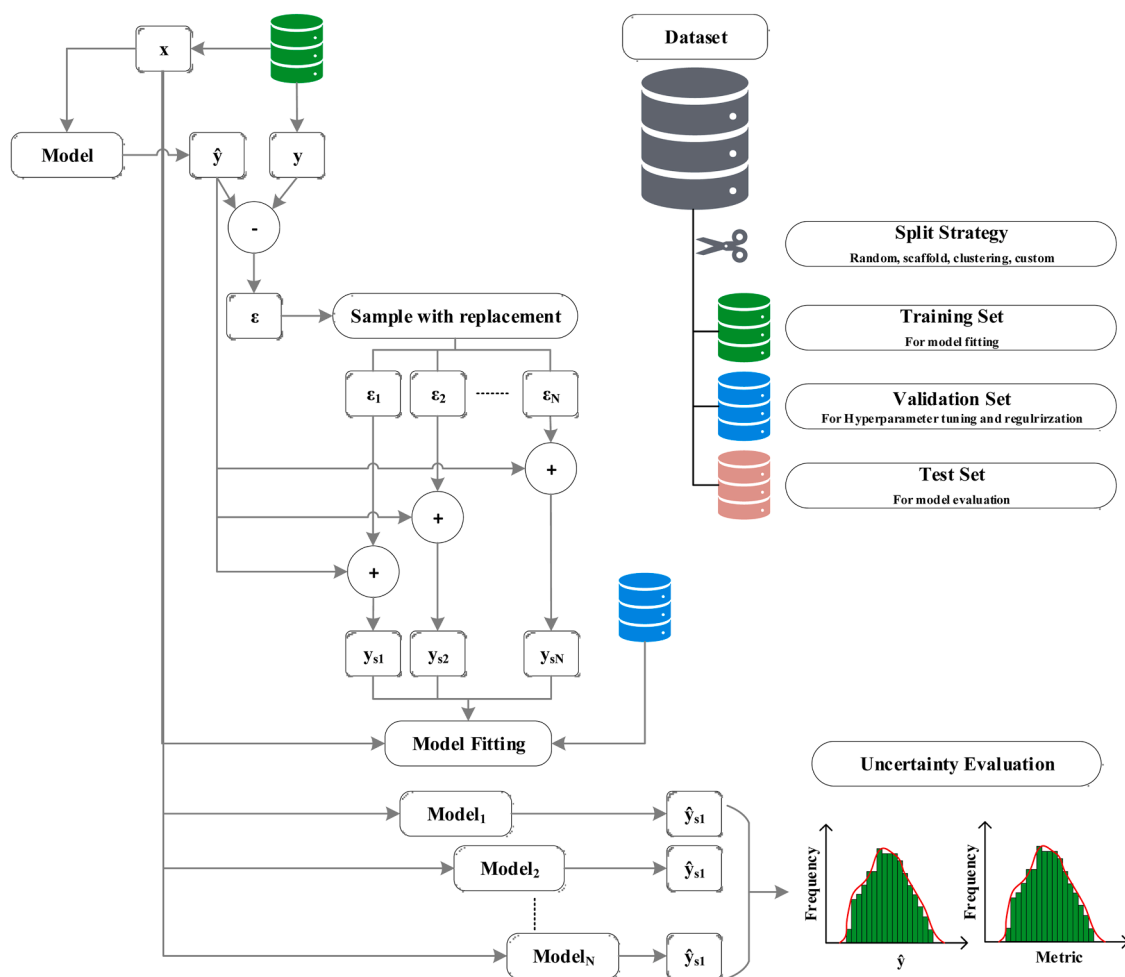


Fig. 7. Overview of the bootstrap technique used for evaluating the uncertainty in model prediction.

An overview of the various modelling techniques can be seen in Fig. 6.

2.4.1. Tree-based models

A decision tree (DT) regressor aims to learn simple decision rules from the provided features. As such, the model presents a series of splits (or decisions) regarding the features provided and segments the prediction space into segments relating to the feature splits (Breiman, 2017). These decisions are then adjusted based on a loss function, e.g., mean squared error. Random forests (RF) are an ensemble technique that leverages multiple decision trees and provides a prediction that is an aggregation of the individual “weak learners” (Breiman, 2001). Each decision tree is trained on samples drawn (with replacements) from the total available training set. Gradient boosted trees (GB) is another ensemble technique that trains a series of sequential DT models that aim to model the error resulting from the previous model (Friedman, 2001). Finally, extreme gradient-boosted trees, also referred to as XGBoost, are considered. These models come with a series of improvements compared to GB in terms of regularization and parallelization (Chen and Guestrin, 2016).

2.4.2. Support vector regressors

Support Vector Machine (SVM) models originated as a binary classification algorithm but were subsequently generalized to handle regression problems with continuous target variables. The core mechanism of SVMs involves constructing an optimal hyperplane that maximizes the margin between classes in the feature space (Cortes and Vapnik, 1995). This decision boundary is mathematically defined as a

Table 2

Performance of GCM using various splitting methods showing the number of compounds considered in the testing as well as the coefficient of determination (R^2) and the mean absolute error (MAE).

Dataset	Metric	Split type				
		No split	Random	Butina	Minimal coverage	Minimal coverage + Butina
T_c [K]	No.	1,057	159	158	841	159
	R^2	0.96	0.88	0.86	0.70	0.95
	MAE	11.35	19.25	19.63	36.10	13.97
P_c [bar]	No.	912	137	137	706	137
	R^2	0.96	0.55	0.86	0.78	0.91
	MAE	1.32	3.11	2.38	2.44	2.12
V_c [L/mol]	No.	783	118	118	590	118
	R^2	0.99	0.01	0.01	0.96	0.99
	MAE	0.013	0.95	0.280	0.019	0.019
ω [-]	No.	480	72	72	324	72
	R^2	0.91	0.54	0.52	0.072	0.91
	MAE	0.027	0.06	0.065	0.136	0.036

No split: the metrics reported are when all data is used for the model calibration, as traditional GCM methods use.

For random, Butina and the hybrid combination of minimal coverage and Butina clustering, the metrics are evaluated on a test set covering 15% of the total dataset. For minimal coverage, the metric is reported for all data not considered a part of the minimal covering set.

For the random split, the average is reported over the three random splits.

Table 3

Loss of groups based on the various splitting strategies across the properties.

Dataset	Group order	Total	Split types			
			Random	Butina	Minimal covering	Minimal covering + Butina
T _c [K]	1 st	122	[-9, -6, -8]	-7	0	0
	2 nd	73	[-10, -8, -12]	-6	0	0
	3 rd	27	[-1, -4, -2]	-7	0	0
P _c [bar]	1 st	118	[-4, -8, -7]	-6	0	0
	2 nd	69	[-10, -9, -10]	-5	0	0
	3 rd	26	[-5, -3, -8]	-5	0	0
V _c [L/mol]	1 st	114	[-13, -8, -4]	-9	0	0
	2 nd	65	[-7, -9, -8]	-12	0	0
	3 rd	22	[-2, -4, -6]	-3	0	0
ω [-]	1 st	104	[-13, -11, -12]	-8	0	0
	2 nd	46	[-8, -4, -6]	-7	0	0
	3 rd	15	[-3, -4, -1]	-2	0	0

No split: the metrics reported are when all data is used for the model calibration, as traditional GCM methods use.

For random, Butina and the hybrid combination of minimal coverage and Butina clustering, the metrics are evaluated on a test set covering 15% of the total dataset.

For minimal coverage, the metric is reported for all data not considered a part of the minimal covering set.

For the random split, we report the missing groups for each of the splits, these will change with each random seed.

Table 4

Performance of GCM based on the hybrid split (minimal covering + Butina) compared to previously published models using the same groups. For the GCM developed in this study, we report the metrics on the test set.

Dataset	Metric	This study	Hukkerikar et al. (2012b)	Aouichaoui et al. (2022a)
T _c [K]	No.	159	858	817
	R ²	0.95	0.99	0.99
	MAE	13.97	7.72	4.57
P _c [bar]	No.	137	852	811
	R ²	0.91	0.97	0.99
	MAE	2.12	1.40	0.74
V _c [L/mol]	No.	118	797	761
	R ²	0.99	0.99	0.99
	MAE	0.019	0.008	0.007
ω [-]	No.	72	1,723	1,636
	R ²	0.91	0.89	0.97
	MAE	0.036	0.053	0.038

linear combination of kernel functions, which map the input data into a higher-dimensional space where linear separation becomes possible. The hyperplane is parameterized by support vectors - the critical data points that determine the position and orientation of the decision boundary.

2.4.3. Gaussian process regressors

Gaussian Process Regression (GPR) is a probabilistic, non-parametric approach that models a function by defining a prior distribution over all possible functions that could fit the data. Rather than learning specific parameters, GPR learns the statistical relationships between points using kernel functions (also called covariance functions). These kernels determine how strongly the function values, at different points, influence each other based on their distance and characteristics (Rasmussen and Williams, 2006).

The key feature of GPRs is that they provide not just predictions but also uncertainty estimates, expressing their confidence in different regions of the input space. When making predictions, GPR computes a posterior distribution that combines the prior beliefs with observed data, yielding both a mean prediction and variance at each point.

2.4.4. Multi-layer perceptron (MLP)

The multi-layer perceptron is a deep feed-forward neural network characterized by a depth (number of hidden layers), a width (number of neurons in each layer) and a non-linear activation function to introduce non-linearity into the model. Each of the nodes is connected to the nodes in the previous layers by a set of weights that are automatically adjusted using backpropagation of the errors (Rumelhart et al., 1986). These models operate under the universal approximation theory and can, in theory, approximate any continuous non-linear function given enough

data and a suitable architecture (Hornik et al., 1989).

2.5. GNNs

Graph neural networks (GNNs) are comprised of two linked sub-processes: a feature extraction process (through the message-passing framework) and a regressor process through an MLP. In this work, three GNNs are considered: the MatERials Graph Network (MEGNet) by Chen et al. (2019), the attentiveFP by Xiong et al. (2020) and the GroupGAT by Aouichaoui et al. (2023c). It is important to note that the MEGNet in this work was modified to not include global features relating to the molecule or the system. Otherwise, the main difference between the models resides in the operations chosen for the message-passing framework. For the mathematical formulas of each model, the reader is referred to the original publications. The chemical information used to featurize the molecular graph is described in the supplementary material.

2.6. Uncertainty estimation

Despite numerous publications highlighting the necessity of providing the uncertainties associated with the model prediction, an increasing number of studies still omit such analysis. While considered too challenging in the past, numerous studies have proposed a wide range of techniques for uncertainty estimation of ML-based models such as: the Monte Carlo dropout (Gal and Ghahramani, 2016) and various ensemble techniques (weight initialization, bootstrap aggregation (Hirschfeld et al., 2020), Bayesian bootstrap (Rubin, 1981)). In this work, some models, such as GPRs, have an inherent ability to provide the uncertainty associated with the prediction. However, to provide the

Table 5

Performance of GCM, GC-ML and the various GNNs across the dataset on the external test set.

Model	T_c [K]				P_c [bar]				V_c [L/mol]				ω [-]			
	R^2	RMSE	MAPE	MAE	R^2	RMSE	MAPE	MAE	R^2	RMSE	MAPE	MAE	R^2	RMSE	MAPE	MAE
GCM	0.95	24.46	2.62	13.97	0.91	3.89	6.00	2.12	0.99	0.03	3.42	0.02	0.91	0.05	11.75	0.04
GC-DT	0.65	65.32	8.42	46.97	0.60	8.12	19.43	6.20	0.90	0.10	16.37	0.07	0.66	0.10	23.71	0.07
GC-RF	0.86	41.63	5.81	31.98	0.82	5.43	9.68	3.14	0.88	0.11	12.57	0.07	0.86	0.06	12.95	0.05
GC-GB	0.91	32.75	3.90	21.39	0.88	4.46	8.12	2.72	0.96	0.06	7.77	0.04	0.85	0.06	11.84	0.04
GC-XGB	0.90	34.11	4.13	22.94	0.85	4.94	9.18	3.04	0.92	0.09	8.51	0.05	0.90	0.05	12.47	0.04
GC-GPR	0.94	27.10	2.43	13.42	0.88	4.55	5.71	2.15	0.99	0.03	2.84	0.02	0.90	0.05	10.97	0.03
GC-SVR	0.94	27.74	2.60	14.64	0.86	4.82	8.75	2.36	0.99	0.04	3.37	0.02	0.92	0.05	9.99	0.03
GC-MLP	0.95	24.03	2.64	14.39	0.89	4.33	7.60	2.55	0.98	0.04	6.07	0.03	0.93	0.05	11.84	0.03
MEGNet	0.92	31.93	3.69	20.75	0.85	5.00	8.35	2.56	0.92	0.09	9.61	0.05	0.82	0.07	14.46	0.05
attentiveFP	0.95	24.24	2.17	12.37	0.86	4.82	5.60	2.02	0.99	0.03	3.21	0.02	0.94	0.04	10.16	0.03
GroupGAT	0.97	19.53	1.75	9.78	0.88	4.42	5.65	1.91	0.99	0.03	2.86	0.01	0.94	0.06	9.80	0.03

same basis for comparison, this study uses the bootstrap method proposed by Efron (1992). This method has previously been used in studies relating to the uncertainty estimation of GCM (Frutiger et al., 2016b) as well as benchmark studies for various uncertainty estimation techniques applied to GNNs for property estimation (Aouichaoui et al., 2022b). The method consists of establishing a reference model, after which the prediction errors are calculated. N synthetic datasets (in this work $N=100$) are constructed by sampling with replacements from the errors and adding the samples to the reference prediction. The model is then retrained on the new synthetic dataset, and the model parameters are saved. This will result in an ensemble of N model realizations, which can then be used to generate different predictions for the same input. An overview of the bootstrap method is shown in Fig. 7.

2.7. Training

For the GC-ML and GNN, the model weights are calibrated using the training set, while the validation set is used to optimize the various hyperparameters associated with the model. Meanwhile, the test is used as an external validation set for the final model (see Fig. 7 for the intended use of each set).

The GCM does not require any hyperparameter tuning, the training set was used for model calibration only. The validation and test sets are considered two different external validation sets. For GC-ML and the GNNs, the target output was scaled using the Z-score scaling. The input features for GC-ML are kept as is, in integer format, as the models seemed to be insensitive to their scaling as well as to retain a physically meaningful input format.

Each model for each dataset has undergone hyperparameter optimization. The ML-based models (DT, RF, SVR, GPR, GB, XGB) and the GNN and MLP models were trained using the Optuna Python package (Akiba et al., 2019). Tree-structured Parzen estimator (TPE) algorithm was used for the hyperparameter optimization (Bergstra et al., 2011; Watanabe, 2023). TPE works by modelling successful and unsuccessful parameter combinations separately. For each parameter, it creates two probability distributions using Gaussian Mixture Models (GMMs). The first GMM, $l(x)$, represents the parameter values that led to the best performance. The second GMM, $g(x)$, represents all other parameter values that were less successful. TPE then selects new parameter values by finding points where the ratio $l(x)/g(x)$ is highest - in other words, where there is the greatest probability of finding good parameters relative to bad ones. This approach efficiently guides the search toward promising regions of the parameter space. An overview of the ranges of the parameters considered and the tuning settings can be found in the supplementary material.

The ML-based models (DT, RF, SVR, GPR, GB, XGB) were trained on an Intel Xeon Gold 6226R. The MLP models were trained on an AMD EPYC 9354. Meanwhile, the GNN models were trained on an Nvidia Volta 100 GPU. All computing resources were available through the DTU computing center (DCC) (DTU Computing Center, 2024).

3. Results and discussion

3.1. The effect of data splitting on the performance of GCM

One major issue with the development of prior GCMs is that many studies did not consider an external test set to demonstrate the wider applicability of the models and to align the development procedure of GCMs with that of various ML-based models. Table 2 showcases the performance of the GCM developed based on 4 different splitting strategies: random split (3 repetitions), butina-based split, minimal coverage split and a hybrid approach combining the minimal covering split and butina-based split. To highlight the possible dangers of using all data for model calibration, the performance metric of a model fitted on all data is also shown in Table 2. As can be seen from Table 2, it might not be appropriate to “just plug-in” groups in an ML-based workflow. This is clear from the fact that the performance of the GCMs, degrades significantly when using typical ML splits. Traditional GCM workflow using all data (no split) for model calibration would result in the “highest” achievable accuracy. When compared to split strategies such as random splitting and Butina-based splitting, it is clear that the performance of the model degrades significantly for all properties. In comparison, the proposed hybrid split (minimal coverage + Butina), consistently provides performances close to the highest achievable accuracies despite using the same amount of data. This is not surprising considering the loss of information resulting from random and Butina-based splitting as seen in Table 3. The effect of random splitting on this imbalanced data can be rather severe, with losses close to 20% of available groups. This can easily lead to numerical issues when performing the parameter fitting, as some predictions can exhibit huge deviations and some severe outliers (as can be seen in Table 2 and from the value of R^2). It is also important to note that the minimal coverage split alone is no guarantee of a good model performance (except V_c). While the algorithm does define the minimal set of molecules exhibiting all available groups, the reliability of the choice of molecule is not taken into account. This is largely due to a lack of designated experimental uncertainties, which could have been integrated to rank potential choices of molecules to include in the minimal covering set.

Furthermore, defining a minimal set of compounds exhibiting all groups could result in identifiability issues in the model calibration. In fact, this has been previously observed even when using all available data (Frutiger et al., 2016b). This could provide a reason for the decrease in performance, especially for the acentric factor. Therefore, it is important to either supply the training split with additional examples (e.g. minimal coverage + Butina) or to exclusively use reliable data. The latter would require expert knowledge capable of evaluating the confidence in the published data. Adhering to the norm in GC modelling and using all data for model calibration does not align with modern ML-based workflows. Not only would this result in overfitted ML models, it also provides GCMs an unfair advantage when compared with ML models. This is especially true for properties where the past few

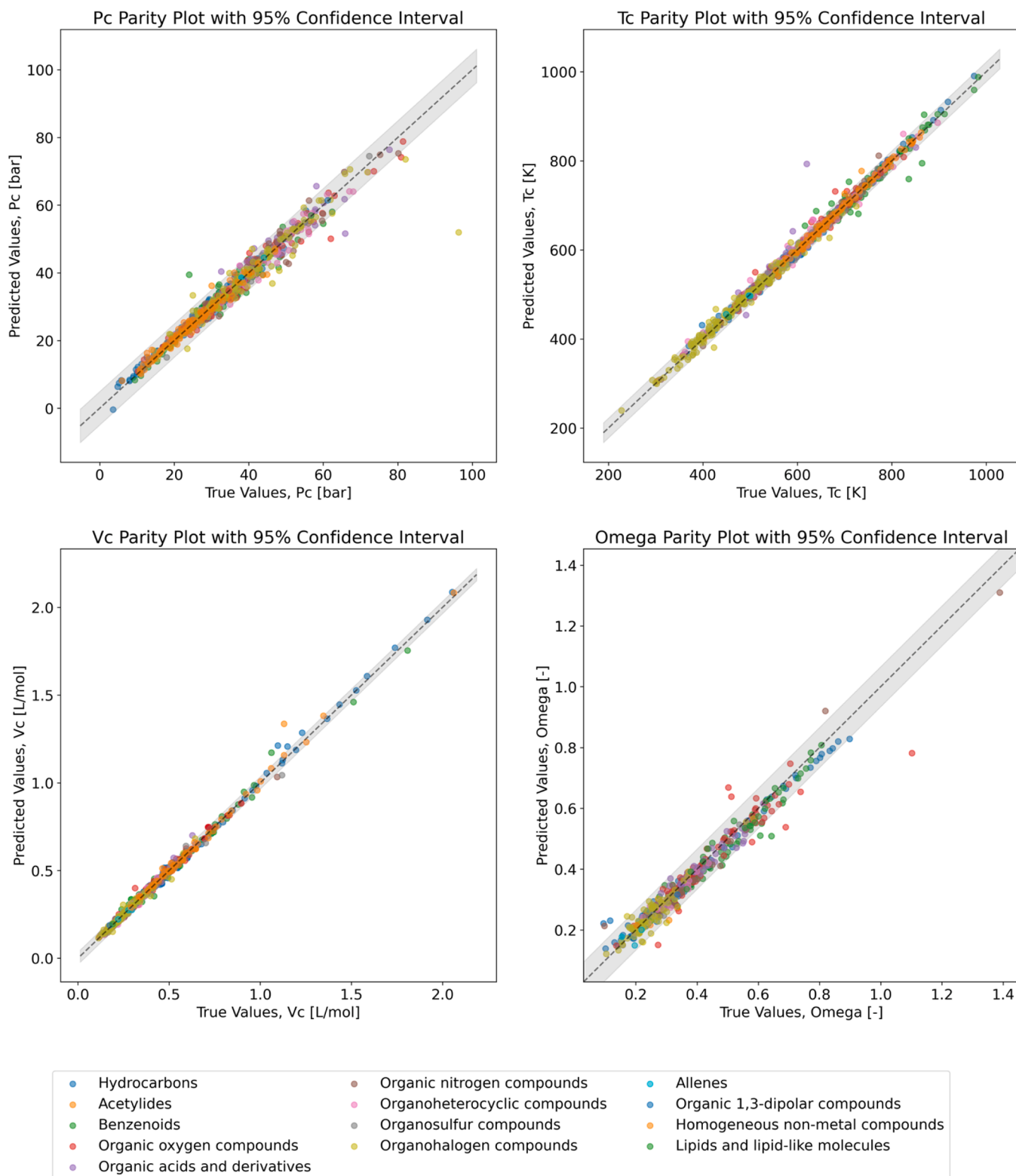


Fig. 8. Parity plot of the GroupGAT model across all datasets for the chemical species present in the data. The shaded area represents the 95% confidence interval for model predictions.

years have only brought small incremental improvements. As such, an overly optimistic performance of the models produced by using all data for regression would make it difficult to truly assess, current advancement in using ML approaches.

3.2. Performance of GCM compared to similar GCM methods

We compare the obtained models with that of [Hukkerikar et al. \(2012b\)](#) and [Aouichaoui et al. \(2022a\)](#). The main motivation is that all studies employ the same group-contribution method (Marrero-Gani). The main difference between those two studies is the application of

Table 6

Performance of GCM, GC-ML and GroupGAT on the test set generated from the Butina-based split.

Properties	Metric	Model		
		GCM	GC-ML	GroupGAT
T_c [K]	R^2	0.86	0.93	0.98
	MAE	19.63	22.37	10.96
P_c [bar]	R^2	0.86	0.86	0.95
	MAE	2.38	2.86	1.78
V_c [L/mol]	R^2	0.01	0.96	0.98
	MAE	0.280	0.022	0.015
ω [-]	R^2	0.52	0.81	0.93
	MAE	0.065	0.050	0.030

Butina-based split is used for the analysis.

outlier detection as described by [Frutiger et al., \(2016b\)](#), as well as the source of data. This explains why [Aouichaoui et al. \(2022a\)](#) used only a subset of the data from [Hukkerikar et al. \(2012b\)](#) as shown in [Aouichaoui et al. \(2022a\)](#) which excluded outliers. Outliers here are defined as data points that are outside the critical threshold of 2.5% of the empirical cumulative distribution function of the error between the model prediction and target value. This does not necessarily reflect actual experimental/measurement errors but rather outliers based on the mismatch between a model structure and data.

When compared to the study by [Hukkerikar et al. \(2012b\)](#), as shown in [Table 4](#), the current GCM model provides better metrics for the acentric factor. While the improvements can be associated with the use of a smaller dataset, it is important to highlight the potential use of predicted values rather than experimentally derived values in the study by [Hukkerikar et al. \(2012b\)](#). This is especially visible for the acentric

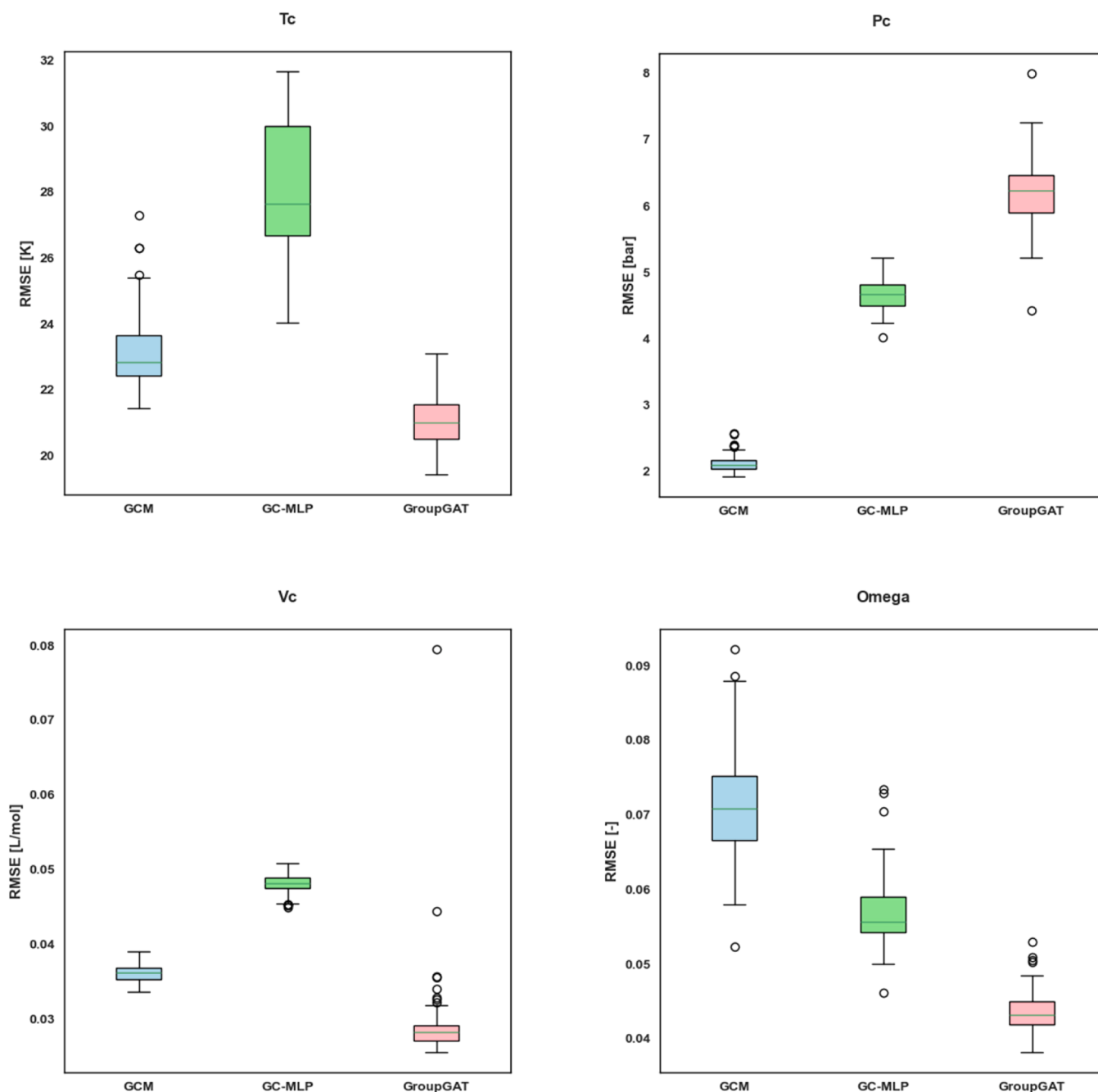


Fig. 9. Boxplot of the variance registered across bootstrap samples with respect to the root mean squared error (RMSE) for GCM, GC-MLP and GroupGAT.

Table 7

Coverage of the 95% confidence interval and the mean standard deviation (σ) of the predicted samples generated from the bootstrap method to the mean in addition the number of trainable parameters N.

	T_c [K]			P_c [bar]			V_c [L/mol]			ω [-]		
	GCM	GC-MLP	GroupGAT	GCM	GC-MLP	GroupGAT	GCM	GC-MLP	GroupGAT	GCM	GC-MLP	GroupGAT
Coverage	57.99	71.33	70.39	59.98	75.11	86.84	73.82	68.71	78.54	82.71	77.30	56.46
σ	6.745	8.121	4.199	0.836	1.016	2.194	0.009	0.010	0.01	0.028	0.020	0.009
N	222	48,641	6,490,385	213	104,849	2,576,737	201	32,065	1,228,041	165	168,241	2,869,577

factor, which is derived from knowledge about the vapor pressure and the critical temperature through the Pitzer correlation and is thus ultimately limited by the availability of both as seen in Table 4.

While the current GCM shows lower performance metrics compared with the previous models in terms of raw performance for T_c and V_c , it is important to note that the current model is built on a fairer foundation of data splitting for training and testing. This ensures a more reliable and unbiased evaluation of the predictive capabilities of the model. Another point to highlight is that, while outlier detection does improve the model performance statistics, this is not surprising as it simply omits extreme mismatching observations from the sample data. However, this comes at the expense of reducing the size of the chemical compounds predicted successfully by the model. In short, this procedure is purely statistical and is not recommended as it does not rely on any chemical prior knowledge nor any information related to the measurement error associated with the given compounds.

3.3. Effect of ML on GC-based models

The performance of GC-ML differs greatly depending on the choice of ML algorithm, with many falling below the linear GCMs as seen in Table 5. In this analysis, we focus on the MAE metric, and if these are equal, the RMSE is used, and if these are equal, the MAPE is used. GCMs are only outperformed by GC-GPR for the T_c and V_c , and GC-GPR, GC-SVR and GC-MLP for ω . Meanwhile, none of the GC-MLs outperform the GCM for the P_c . While this might be surprising at first, one explanation could be the limiting value integrated into the GCM of $P_{c\infty} = 0.0519$ bar (Tsonopoulos and Tan, 1993). This would provide better predictions, especially for compounds with lower P_c values. Such knowledge is not easily incorporated into the GC-MLs and must be inferred from the data. However, it is expected that very few compounds exhibit such values and as such, this exposes the potential limits of GC-ML. An important takeaway from comparing GCMs and GC-MLs is that the inclusion of ML does not always yield better results.

When comparing the GC-MLs internally, not surprisingly, DTs perform the worst across all datasets. This is because, given the large number of features used, many decision boundaries must be constructed to achieve high accuracy. As such, DTs can be seen as a coarse form of regression. RFs consistently outperform the DTs, which is expected since RFs are an ensemble of DTs and can create a better set of decision boundaries. GB and XGB also outperform RFs, which is expected since they construct an ensemble that continuously performs error modelling. The performance of GB and XGB is very comparable, with GB marginally outperforming XGB. This could be due to the regulations enforced in XGB. GC-GPR and GC-SVR perform comparably across all datasets, along with GC-MLP. This is slightly surprising given that the MLP should be capable of approximating any function (Hornik et al., 1989). It is, however, important to note the computational challenge in training MLP rather than SVRs and GPRs could explain the difference in performance.

3.4. GCM, GC-ML or GNN?

As shown in Table 5, all GNNs except MEGNet outperform GCM and GC-ML. One major reason could be the modification imposed on MEGNet as it does not consider any global molecule-level features. This was one of the strengths of the MEGNet (the omission of the spacial distances

and other global features) as illustrated by Chen et al. (2019). GroupGAT outperforms the attentiveFP model across all properties. This is not surprising as the GroupGAT is composed of three different attentive FP models to encode the hierarchical structural information characterizing the GroupGAT model. The MAE is improved by 30%, 10%, 20% and 15% for T_c , P_c , V_c and ω , respectively when compared to GCM. The parity plot for the GroupGAT across all datasets can be seen in Fig. 8. It highlights the various classes of compounds, along with the associated uncertainty according to a t-student distribution with a critical value of 1.96. For T_c , solely 36 (app. 3.4%) datapoints are outside the calculated confidence interval. Furthermore, it is clear that two compounds are well beyond that confidence interval and the model fails to predict these. The first compound is maleic acid (part of the test set). While it is not entirely clear why this compound is predicted badly, it is most possibly due to a lack of features to describe the isomeric structure as previously highlighted in Aouichaoui et al. (2023c). The other compound is isoquinoline. Since the GroupGAT relies on the groups defined by the first-order groups of Marrero-Gani for two of the attentiveFP models it is made of, it is important to note that the ring/pyridine fusion is described by the second-order groups. As such, this would potentially explain the discrepancy associated with the compound. For P_c , the compounds outside of the confidence interval account for 4.7% (app. 43 compounds) and exhibit more spread and variety in terms of classes of compounds. The same goes for V_c , where 43 compounds are beyond the confidence interval (5.5%). Finally, for ω , the model exhibits a few more noticeable outliers, most of which are halogenated organics and oxygen-containing compounds. These compounds cover tetrabromomethane, propylene glycol and diisopropanolamine.

3.5. Robustness of GC-ML and GNNs toward missing groups

The impact of missing groups resulting from data splitting has already been highlighted previously in Table 2 for the GCM. To highlight the robustness of GroupGAT and GC-MLP are trained on the Butina-cluster based split. The model performance is shown in Table 6. As can be seen, it seems that the GC-ML and GroupGAT are much more robust against missing groups than the GCM. GroupGAT retains its excellent performance despite it being based on the substructures as the first order groups used in GCM. However, these groups are featurized using smaller entities in the form of atoms and bonds and are less sensitive to the absence of the specific arrangement of the atoms.

3.6. Uncertainty estimation

The bootstrap method, as previously described and illustrated in Fig. 7, generates new synthetic data by sampling with replacement the residuals between the ground truth (target) and a reference model. In this work, 100 bootstrap samples were generated, and for each sample, the models were retrained and their metrics quantified. Note that synthetic data was only generated for the training data, and the model evaluation was still based on the ground truth of the validation and test sets. The uncertainty estimation is evaluated using two metrics: the coverage and the mean standard deviation (σ) of the error between the test set ground truth and the bootstrap predictions. The coverage is defined as the percentage of measurement points that fall within the range of the 95%-confidence interval (CI) (Kompa et al., 2021). The CI is

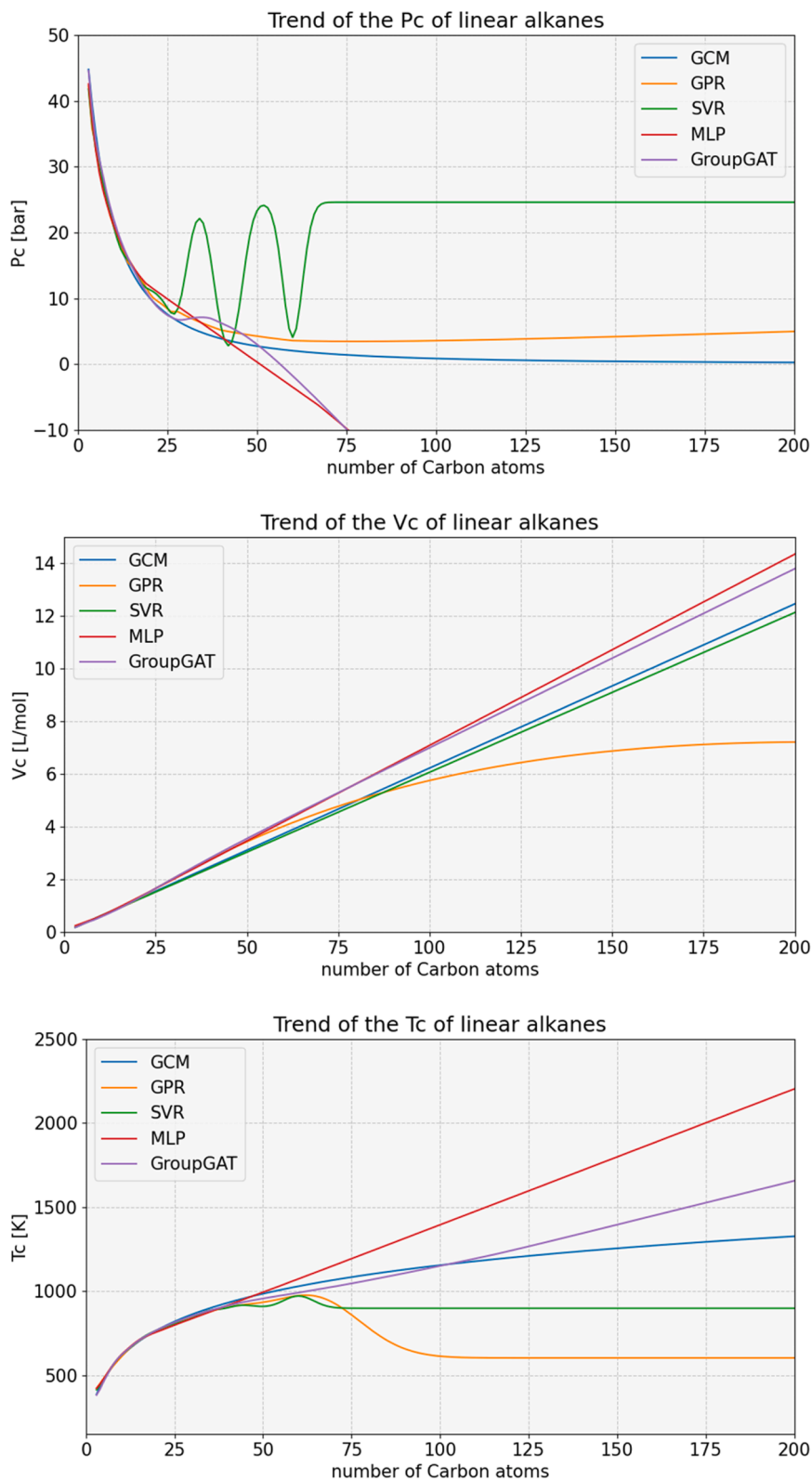


Fig. 10. Trend of the P_c , V_c and T_c for linear alkanes up to 200 Carbon atoms.

calculated as the quantile of the generated sample distribution. No method consistently outperforms another. Of note is that the evaluation of the uncertainty should take into account the trade-off between width (in the form of σ) and coverage of the CIs, as highlighted in previous studies (Aouichaoui et al., 2022b). The desired scenario is to obtain a low σ and a high coverage. The main reason is that complete coverage

can be achieved if the CI is wide enough, however, this renders the prediction rather unreliable and would pose potential issues for applications such as Monte-Carlo simulations of various processes in which the properties are used.

Fig. 9, shows the ensemble of bootstrap accuracy as measured by root mean squared error (RMSE). The variance shown in the Fig. 9 indicates

how the RMSE score is influenced/varied by the particular training data randomly sampled in the bootstrap. Of interest is the mean RMSE (the middle line in the box chart), which is an indication of model accuracy averaged-out across all the data in the test set. We make the following observations: (i) GroupGAT provides more accurate predictions (indicated by lower RMSE scores) for the T_c , V_c and ω , and (ii) interestingly, for P_c property, GC-MLP and GCM models achieve better RMSE scores (meaning higher accuracy), despite GroupGAT provided a higher coverage (see Table 7). The conclusion here is that for most properties, GroupGAT provided generally more accurate and high coverage for ground truth considered in the test datasets.

3.7. Consistency analysis

As a final element of the study, we perform a consistency check of the models by observing the trends of T_c , P_c , V_c and the calculated critical compressibility factor Z_c for linear alkane chains (up to C=200), similarly to the studies made by Hukkerikar et al. (2012b). Z_c is calculated according to Eq. (7), where R is the universal gas constant.

$$Z_c = \frac{P_c \cdot V_c}{R \cdot T_c} \quad (7)$$

The trends have been constructed for Alkanes up to 200 Carbon atoms and are shown in Fig. 10. As can be seen, all ML-based methods start deviating around a chain length of 25, with some models exhibiting erratic behavior (especially GPR, GC-MLP and GroupGAT). None of the models reaches the limiting value reported by Tsonopoulos and Tan (1993). This shows that ML-based models can be limited in terms of their ability to extrapolate beyond their domain of applicability. This is not the case for the GCM, where the limiting value can be embedded as in Eq. (3) in the regression constants. Important to note is also that the DL-based methods breach the physics by providing negative values for P_c . While not the focus of this study, this can be remedied by clipping and setting constraints for the model outputs.

For V_c , all models exhibit a linear trend, which is also expected from the property. The main difference is the slope and the fact that for GC-SVR, a horizontal asymptotic value is reached, which corresponds to the mean value of the dataset. For T_c , and similar to P_c , the models start diverging around alkanes of chain length 50, with MLP and GroupGAT increasing linearly, while GC-GPR exhibits a decrease and a convergence towards the mean value of the dataset. Considering the behavior with respect to T_c , P_c and V_c , it is clear then that the ML-based methods would not present the limiting value suggested for the Z_c .

3.8. Limitations and future perspectives

As previously highlighted, the selection of the minimal covering set is dependent on the ordering of the molecules and does not take into account the quality of the data. Therefore, we suggested supplementing the minimal covering set with a subset of the data using either random sampling or Butina-based split. Awareness of the quality of the data used for model calibration is important and can influence the final model. Because few databases provide the associated uncertainties, it is important to emphasize this in future data collection (Bazyleva et al., 2021). An alternative approach that could serve a similar purpose is the use of active learning to select the required subset of the data to increase the robustness of the prediction and reduce the variance in accuracy (Liu et al., 2024; Smith et al., 2018).

4. Conclusions

Group Contribution models (GCM) for chemical property prediction face ongoing scrutiny regarding the use of all available data for model calibration, which not only hinders the comparison with ML-based methods but also results in models whose extrapolative ability remains unverified. However, direct integration of models using groups as

features will result in limiting their domain of applicability by potentially losing available groups during the data splitting. In this study, we not only showcase this deterioration of performance from missing groups but also propose a hybrid splitting scheme that combines the concept of minimal coverage (identifying the smallest subset of compounds that feature all available groups) with the cluster-based Butina split method. This method provides performance close to that of using all available data with only a fraction of the data, providing a rigorous and fairer foundation for their comparison with other ML models. By benchmarking various split techniques, we showcased that ML-based methods are more robust to missing features, even when having groups missing compared to the traditional GCM. Utilizing the novel split techniques, we further showcase that ML-based methods outperform classical GCM in terms of performance metrics, with GroupGAT (a graph neural network) being the best-performing model. However, as highlighted, it is important to use these models with care when extrapolating beyond the domain of applicability, as these exhibit strange trends of the critical properties for linear alkanes with chain lengths above 25. Finally, in terms of uncertainties, the bootstrap method showed that no method outperforms the other in terms of coverage and width of the uncertainty, expressed through the mean standard deviation of the predictions.

Data and code availability

The code and data for generating the results presented in this work are made available on GitHub: https://github.com/arnaou/fair_gc. The newly calibrated group contributions are also made available along with the trained machine learning models.

Declaration of generative AI in scientific writing

During the preparation of this work, the author(s) used Claude.ai by Anthropic to improve the readability and language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

CRediT authorship contribution statement

Adem R.N. Aouichaoui: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jingkang Liang:** Writing – review & editing, Visualization, Software, Methodology, Formal analysis. **Jens Abildskov:** Writing – review & editing, Supervision, Conceptualization. **Gürkan Sin:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Gürkan Sin reports financial support was provided by EU Horizon 2020 Framework Programme. Jingkang Liang reports financial support was provided by EU Horizon 2020 Framework Programme. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.compchemeng.2025.109271](https://doi.org/10.1016/j.compchemeng.2025.109271).

References

- Akiba, T., Sano, S., Yanase, T., Ohta, T., Koyama, M., 2019. Optuna: a next-generation hyperparameter optimization framework. In: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*.
- Al, R., Frutiger, J., Zubov, A., Sin, G., 2018. Prediction of environmental properties using a hybrid group contribution approach. *Computer Aided Chemical Engineering*. Elsevier, pp. 1723–1728. <https://doi.org/10.1016/B978-0-444-64241-7.50282-2>.
- Alshehri, A.S., Tula, A.K., You, F., Gani, R., 2022. Next generation pure component property estimation models: with and without machine learning techniques. *AIChE J.* 68, e17469. <https://doi.org/10.1002/aic.17469>.
- Aouichaoui, A.R.N., Al, R., Abildskov, J., Sin, G., 2021. Comparison of group-contribution and machine learning-based property prediction models with uncertainty quantification. In: *Türkay, M., Gani, R. (Eds.), Computer Aided Chemical Engineering, 31 European Symposium on Computer Aided Process Engineering*. Elsevier, pp. 755–760. <https://doi.org/10.1016/B978-0-323-88506-5.50118-2>.
- Aouichaoui, A.R.N., Mansouri, S.S., Abildskov, J., Sin, G., 2022a. Application of outlier treatment towards improved property prediction models. In: *Montastruc, L., Negny, S. (Eds.), Computer Aided Chemical Engineering, 32 European Symposium on Computer Aided Process Engineering*. Elsevier, pp. 1357–1362. <https://doi.org/10.1016/B978-0-323-95879-0.50227-7>.
- Aouichaoui, A.R.N., Mansouri, S.S., Abildskov, J., Sin, G., 2022b. Uncertainty estimation in deep learning-based property models: graph neural networks applied to the critical properties. *AIChE J.* 68, e17696. <https://doi.org/10.1002/aic.17696>.
- Aouichaoui, A.R.N., Cogliati, A., Abildskov, J., Sin, G., 2023a. S-GNN: state-dependent graph neural networks for functional molecular properties. In: *Kokossis, A.C., Georgiadis, M.C., Pistikopoulos, E. (Eds.), Computer Aided Chemical Engineering, 33 European Symposium on Computer Aided Process Engineering*. Elsevier, pp. 575–581. <https://doi.org/10.1016/B978-0-443-15274-0.50091-3>.
- Aouichaoui, A.R.N., Fan, F., Abildskov, J., Sin, G., 2023b. Application of interpretable group-embedded graph neural networks for pure compound properties. *Comput. Chem. Eng.* 176, 108291. <https://doi.org/10.1016/j.compchemeng.2023.108291>.
- Aouichaoui, A.R.N., Fan, F., Mansouri, S.S., Abildskov, J., Sin, G., 2023c. Combining group-contribution concept and graph neural networks toward interpretable molecular property models. *J. Chem. Inf. Model.* 63, 725–744. <https://doi.org/10.1021/acs.jcim.2c01091>.
- Bazyleva, A., Abildskov, J., Anderko, A., Baudouin, O., Chernyak, Y., De Hemptinne, J.-C., Diky, V., Dohrn, R., Elliott, J.R., Jacquemin, J., Jaubert, J.-N., Joback, K.G., Kattner, U.R., Kontogeorgis, G.M., Loria, H., Mathias, P.M., O'Connell, J.P., Schröder, W., Smith, G.J., Soto, A., Wang, S., Weir, R.D., 2021. Good reporting practice for thermophysical and thermochemical property measurements (IUPAC technical report). *Pure Appl. Chem.* 93, 253–272. <https://doi.org/10.1515/pac-2020-0403>.
- Benson, S.W., Buss, J.H., 1958. Additivity rules for the estimation of molecular properties. *Thermodynamic properties*. *J. Chem. Phys.* 29, 546–572. <https://doi.org/10.1063/1.1744539>.
- Bergstra, J., Bardenet, R., Bengio, Y., Kégl, B., 2011. Algorithms for hyper-parameter optimization. In: *Shawe-Taylor, J., Zemel, R., Bartlett, P., Pereira, F., Weinberger, K. Q. (Eds.), Advances in Neural Information Processing Systems*. Curran Associates, Inc.
- Biswas, S., Chung, Y., Ramirez, J., Wu, H., Green, W.H., 2023. Predicting critical properties and acentric factors of fluids using multitask machine learning. *J. Chem. Inf. Model.* 63, 4574–4588. <https://doi.org/10.1021/acs.jcim.3c00546>.
- Breiman, L., 2001. Random forests. *Mach. Learn.* 45, 5–32. <https://doi.org/10.1023/A:1010933404324>.
- Breiman, L., 2017. *Classification and Regression Trees*. Routledge.
- Butina, D., 1999. Unsupervised data base clustering based on daylight's fingerprint and tanimoto similarity: a fast and automated way to cluster small and large data sets. *J. Chem. Inf. Comput. Sci.* 39, 747–750. <https://doi.org/10.1021/ci9803381>.
- Chakraborty, A., Sivaram, A., Venkatasubramanian, V., 2021. AI-DARWIN: a first principles-based model discovery engine using machine learning. *Comput. Chem. Eng.* 154, 107470. <https://doi.org/10.1016/j.compchemeng.2021.107470>.
- Chakraborty, A., Gandhi, A., Hasan, M.M.F., Venkatasubramanian, V., 2024. Discovering zeolite adsorption isotherms: a hybrid AI modeling approach. *Comput. Aided Chem. Eng.* 511–516. <https://doi.org/10.1016/B978-0-443-28824-1.50086-7>.
- Chein-Hsiun, T., 1995. Group-contribution estimation of critical temperature with only chemical structure. *Chem. Eng. Sci.* 50, 3515–3520. [https://doi.org/10.1016/0009-2509\(95\)00191-7](https://doi.org/10.1016/0009-2509(95)00191-7).
- Chen, T., Guestrin, C., 2016. XGBoost: a scalable tree boosting system. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD '16*. New York, NY, USA. Association for Computing Machinery, pp. 785–794. <https://doi.org/10.1145/2939672.2939785>.
- Chen, C., Ye, W., Zuo, Y., Zheng, C., Ong, S.P., 2019. Graph networks as a universal machine learning framework for molecules and crystals. *Chem. Mater.* 31, 3564–3572. <https://doi.org/10.1021/acs.chemmater.9b01294>.
- Constantinou, L., Gani, R., 1994. New group contribution method for estimating properties of pure compounds. *AIChE J.* 40, 1697–1710. <https://doi.org/10.1002/aic.690401011>.
- Corso, G., Stark, H., Jegelka, S., Jaakkola, T., Barzilay, R., 2024. Graph neural networks. *Nat. Rev. Methods Prim.* 4, 17. <https://doi.org/10.1038/s43586-024-00294-7>.
- Cortes, C., Vapnik, V.N., 1995. Support-vector networks. *Mach. Learn.* 20, 273–297.
- Djombou Feunang, Y., Eisner, R., Knox, C., Chepelev, L., Hastings, J., Owen, G., Fahy, E., Steinbeck, C., Subramanian, S., Bolton, E., Greiner, R., Wishart, D.S., 2016. ClassyFire: automated chemical classification with a comprehensive, computable taxonomy. *J. Cheminform.* 8, 61. <https://doi.org/10.1186/s13321-016-0174-y>.
- DTU Computing Center, 2024. DTU Computing Center resources. <https://doi.org/10.48714/DTU.HPC.0001>.
- Efron, B., 1992. Bootstrap methods: another look at the jackknife. In: *Kotz, S., Johnson, N.L. (Eds.), Breakthroughs in Statistics: Methodology and Distribution, Springer Series in Statistics*. Springer, New York, NY, pp. 569–593. https://doi.org/10.1007/978-1-4612-4380-9_41.
- Friedman, J.H., 2001. Greedy function approximation: a gradient boosting machine. *Ann. Statist.* 29. <https://doi.org/10.1214/aos/1013203451>.
- Frutiger, J., Marcarie, C., Abildskov, J., Sin, G., 2016a. Group-contribution based property estimation and uncertainty analysis for flammability-related properties. *J. Hazard. Mater.* 318, 783–793. <https://doi.org/10.1016/j.jhazmat.2016.06.018>.
- Frutiger, J., Marcarie, C., Abildskov, J., Sin, G., 2016b. A comprehensive methodology for development, parameter estimation, and uncertainty analysis of group contribution based property models—an application to the heat of combustion. *J. Chem. Eng. Data* 61, 602–613. <https://doi.org/10.1021/acs.jced.5b00750>.
- Gal, Y., Ghahramani, Z., 2016. Dropout as a bayesian approximation: representing model uncertainty in deep learning. In: *Balcan, M.F., Weinberger, K.Q. (Eds.), Proceedings of The 33rd International Conference on Machine Learning, Proceedings of Machine Learning Research*. New York, New York, USA. PMLR, pp. 1050–1059.
- Gani, R., 2019. Group contribution-based property estimation methods: advances and perspectives. *Curr. Opin. Chem. Eng.* 23, 184–196. <https://doi.org/10.1016/j.coche.2019.04.007>.
- Gharagheizi, F., Mirkhani, S.A., Tofangchi Mahyari, A.R., 2011. Prediction of standard enthalpy of combustion of pure compounds using a very accurate group-contribution-based method. *Energy Fuels* 25, 2651–2654. <https://doi.org/10.1021/ef200081a>.
- Gilmer, J., Schoenholz, S.S., Riley, P.F., Vinyals, O., Dahl, G.E., 2017. Neural message passing for quantum chemistry.
- Haynes, W.M., 2014. *CRC Handbook of Chemistry and Physics*. CRC Press. <https://doi.org/10.1201/b17118>, 0 ed.
- Hirschfeld, L., Swanson, K., Yang, K., Barzilay, R., Coley, C.W., 2020. Uncertainty Quantification using neural networks for molecular property prediction. *J. Chem. Inf. Model.* 60, 3770–3780. <https://doi.org/10.1021/acs.jcim.0c00502>.
- Hornik, K., Stinchcombe, M., White, H., 1989. Multilayer feedforward networks are universal approximators. *Neural Netw.* 2, 359–366. [https://doi.org/10.1016/0893-6080\(89\)90020-8](https://doi.org/10.1016/0893-6080(89)90020-8).
- Hukkerikar, A.S., Kalakul, S., Sarup, B., Young, D.M., Sin, G., Gani, R., 2012a. Estimation of environment-related properties of chemicals for design of sustainable processes: development of group-contribution⁺ (GC⁺) property models and uncertainty analysis. *J. Chem. Inf. Model.* 52, 2823–2839. <https://doi.org/10.1021/ci300350r>.
- Hukkerikar, A.S., Sarup, B., Ten Kate, A., Abildskov, J., Sin, G., Gani, R., 2012b. Group-contribution⁺ (GC⁺) based estimation of properties of pure components: Improved property estimation and uncertainty analysis. *Fluid. Phase Equilib.* 321, 25–43. <https://doi.org/10.1016/j.fluid.2012.02.010>.
- Hukkerikar, A.S., Meier, R.J., Sin, G., Gani, R., 2013. A method to estimate the enthalpy of formation of organic compounds with chemical accuracy. *Fluid. Phase Equilib.* 348, 23–32. <https://doi.org/10.1016/j.fluid.2013.03.018>.
- Jiménez-Luna, J., Grisoni, F., Schneider, G., 2020. Drug discovery with explainable artificial intelligence. *Nat. Mach. Intell.* 2, 573–584. <https://doi.org/10.1038/s42256-020-00236-4>.
- Joback, K.G., Reid, R.C., 1987. Estimation of pure component properties from group-contributions. *Chem. Eng. Commun.* 57, 233–243. <https://doi.org/10.1080/00986448708960487>.
- Kleiber, M., Joh, R., 2010. D3 properties of pure fluid substances. *VDI Heat Atlas*, pp. 301–418. https://doi.org/10.1007/978-3-540-77877-6_18.
- Klincewicz, K.M., Reid, R.C., 1984. Estimation of critical properties with group contribution methods. *AIChE J.* 30, 137–142. <https://doi.org/10.1002/aic.690300119>.
- Kompa, B., Snoek, J., Beam, A., 2021. Empirical frequentist coverage of deep learning uncertainty quantification procedures. *Entropy* 23, 1608. <https://doi.org/10.3390/e23121608>.
- Landrum, G., Tosco, P., Kelley, B., Rodriguez, R., Cosgrove, D., Vianello, R., Sriniker, P. G., Jones, G., NadineSchneider, E.K., Nealschneider, D., Dalke, A., Swain, M., Cole, B., Turk, S., Savelev, A., Vaucher, A., Wójcikowski, M., Take, I., Scalfani, V.F., Walker, R., Probst, D., Ujihara, K., tadhurst-cdd, A.P., Godin, G., Lehtivarjo, J., Bérenger, F., Bisson, J., 2024. rdkit: 2024.09.2 (Q3 2024) Release 10.5281/ZENODO.13990314.
- Lee, C.J., Lee, J.M., 2012. An advanced group contribution method for high-dimensional, sparse data sets. *Mol. Inform.* 31, 41–52. <https://doi.org/10.1002/minf.201100111>.
- Lee, C.J., Ko, J.W., Lee, G., 2012. Flash point prediction of organic compounds using a group contribution and support vector machine. *Korean J. Chem. Eng.* 29, 145–153. <https://doi.org/10.1007/s11814-011-0164-8>.

- Li, R., Herreros, J.M., Tsolakis, A., Yang, W., 2020. Machine learning regression based group contribution method for cetane and octane numbers prediction of pure fuel compounds and mixtures. *Fuel* 280, 118589. <https://doi.org/10.1016/j.fuel.2020.118589>.
- Linstrom, P., 1997. NIST chemistry WebBook. NIST Standard Ref. Datab. 69. <https://doi.org/10.18434/T4D303>.
- Liu, H., Yucel, B., Ganapathysubramanian, B., Kalidindi, S.R., Wheeler, D., Wodo, O., 2024. Active learning for regression of structure-property mapping: the importance of sampling and representation. *Digit. Discov.* 3, 1997–2009. <https://doi.org/10.1039/D4DD00073K>.
- Louis, S.Y., Zhao, Y., Nasiri, A., Wang, X., Song, Y., Liu, F., Hu, J., 2020. Graph convolutional neural networks with global attention for improved materials property prediction. *Phys. Chem. Chem. Phys.* 22, 18141–18148. <https://doi.org/10.1039/D0CP01474E>.
- Mann, V., Venkatasubramanian, V., 2023. AI-driven hypergraph network of organic chemistry: network statistics and applications in reaction classification. *React. Chem. Eng.* 8, 619–635. <https://doi.org/10.1039/D2RE00309K>.
- Marrero, J., Gani, R., 2001. Group-contribution based estimation of pure component properties. *Fluid. Phase Equilib.* 183–184, 183–208. [https://doi.org/10.1016/S0378-3812\(01\)00431-9](https://doi.org/10.1016/S0378-3812(01)00431-9).
- Medina, E.I.S., Linke, S., Stoll, M., Sundmacher, K., 2022. Graph neural networks for the prediction of infinite dilution activity coefficients. *Digit. Discov.* 1, 216–225. <https://doi.org/10.1039/D1DD00037C>.
- Moriwaki, H., Tian, Y.S., Kawashita, N., Takagi, T., 2018. Mordred: a molecular descriptor calculator. *J. Cheminform.* 10, 4. <https://doi.org/10.1186/s13321-018-0258-y>.
- Nannoolal, Y., Rarey, J., Ramjugernath, D., 2007. Estimation of pure component properties: Part 2. Estimation of critical property data by group contribution. *Fluid. Phase Equilib.* 252, 1–27. <https://doi.org/10.1016/j.fluid.2006.11.014>.
- Peng, D., Robinson, D.B., 1977. A rigorous method for predicting the critical properties of multicomponent systems from an equation of state. *AIChE J.* 23, 137–144. <https://doi.org/10.1002/aic.690230202>.
- Perry, R.H., Green, D.W., 2008. *Perry's Chemical Engineers' Handbook*, 8. ed. McGraw-Hill, New York, NY.
- Rasmussen, C.E., Williams, C.K.I., 2006. *Gaussian Processes for Machine Learning, Adaptive Computation and Machine Learning*. MIT Press, Cambridge, Mass.
- Reymond, J.L., 2015. The chemical space project. *Acc. Chem. Res.* 48, 722–730. <https://doi.org/10.1021/ar500432k>.
- Rubin, D.B., 1981. The Bayesian bootstrap. *Ann. Statist.* 9. <https://doi.org/10.1214/aos/1176345338>.
- Ruddigkeit, L., Van Deursen, R., Blum, L.C., Reymond, J.-L., 2012. Enumeration of 166 billion organic small molecules in the chemical universe database GDB-17. *J. Chem. Inf. Model.* 52, 2864–2875. <https://doi.org/10.1021/ci300415d>.
- Rumelhart, D.E., Hinton, G.E., Williams, R.J., 1986. Learning representations by back-propagating errors.
- Simamora, P., Yalkowsky, S.H., 1994. Group contribution methods for predicting the melting points and boiling points of aromatic compounds. *Ind. Eng. Chem. Res.* 33, 1405–1409. <https://doi.org/10.1021/ie00029a042>.
- Smith, J.S., Nebgen, B., Lubbers, N., Isayev, O., Roitberg, A.E., 2018. Less is more: sampling chemical space with active learning. *J. Chem. Phys.* 148, 241733. <https://doi.org/10.1063/1.5023802>.
- Soave, G., 1972. Equilibrium constants from a modified Redlich-Kwong equation of state. *Chem. Eng. Sci.* 27, 1197–1203. [https://doi.org/10.1016/0009-2509\(72\)80096-4](https://doi.org/10.1016/0009-2509(72)80096-4).
- Stefanis, E., Panayiotou, C., 2008. Prediction of hansen solubility parameters with a new group-contribution method. *Int. J. Thermophys.* 29, 568–585. <https://doi.org/10.1007/s10765-008-0415-z>.
- Sun, X., Krakauer, N.J., Politowicz, A., Chen, W., Li, Q., Li, Z., Shao, X., Sunaryo, A., Shen, M., Wang, J., Morgan, D., 2020. Assessing graph-based deep learning models for predicting flash point. *Mol. Inform.* 39, 1900101. <https://doi.org/10.1002/minf.201900101>.
- Tahami, S., Movagharnajad, K., Ghasemtabar, H., 2019. Estimation of the critical constants of organic compounds via a new group contribution method. *Fluid. Phase Equilib.* 494, 45–60. <https://doi.org/10.1016/j.fluid.2019.04.022>.
- Tang, B., Kramer, S.T., Fang, M., Qiu, Y., Wu, Z., Xu, D., 2020. A self-attention based message passing neural network for predicting molecular lipophilicity and aqueous solubility. *J. Cheminform.* 12, 15. <https://doi.org/10.1186/s13321-020-0414-z>.
- Tsonopoulos, C., Tan, Z., 1993. The critical constants of normal alkanes from methane to polyethylene. *Fluid. Phase Equilib.* 83, 127–138. [https://doi.org/10.1016/0378-3812\(93\)87015-S](https://doi.org/10.1016/0378-3812(93)87015-S).
- Watanabe, S., 2023. Tree-structured Parzen estimator: understanding its algorithm components and their roles for better empirical performance. <https://doi.org/10.48550/arXiv.2304.11127>.
- Withnall, M., Lindelöf, E., Engkvist, O., Chen, H., 2020. Building attention and edge message passing neural networks for bioactivity and physical-chemical property prediction. *J. Cheminform.* 12, 1. <https://doi.org/10.1186/s13321-019-0407-y>.
- Xiong, Z., Wang, D., Liu, X., Zhong, F., Wan, X., Li, X., Li, Z., Luo, X., Chen, K., Jiang, H., Zheng, M., 2020. Pushing the boundaries of molecular representation for drug discovery with the graph attention mechanism. *J. Med. Chem.* 63, 8749–8760. <https://doi.org/10.1021/acs.jmedchem.9b00959>.
- Yang, K., Swanson, K., Jin, W., Coley, C., Eiden, P., Gao, H., Guzman-Perez, A., Hopper, T., Kelley, B., Mathea, M., Palmer, A., Settels, V., Jaakkola, T., Jensen, K., Barzilay, R., 2019. Analyzing learned molecular representations for property prediction. *J. Chem. Inf. Model.* 59, 3370–3388. <https://doi.org/10.1021/acs.jcim.9b00237>.
- Yap, C.W., 2011. PaDEL-descriptor: an open source software to calculate molecular descriptors and fingerprints. *J. Comput. Chem.* 32, 1466–1474. <https://doi.org/10.1002/jcc.21707>.
- Yaws, C.L., 2009. *Thermophysical Properties of Chemicals and Hydrocarbons*. William Andrew, Norwich, N.Y.
- Zhang, Z., Guan, J., Zhou, S., 2021. FraGAT: a fragment-oriented multi-scale graph attention model for molecular property prediction. *Bioinformatics* 37, 2981–2987. <https://doi.org/10.1093/bioinformatics/btab195>.